AIM

Application & Infrastructure Monitoring

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1. AIM Overview

One of the main concerns of support teams is the ability to debug quickly and accurately issues opened by customers. Specifically in the cloud infra field the debugging becomes much more complicated considering the complex structure of the infrastructure.
In addition, missing and wrong information sent by the issue opener might cause delays and wrong conclusion of the issue at hand, which causes more time loss.

AIM comes to help cloud-users debug VNFs/CNFs issues from the cloud infrastructure perspective.

AIM gathers all infrastructure information on a specified environment and displays it in a graphic easy to read way. In addition, it gives the user to ability to see the virtual resources used by the users as well.

AIM can collect statistics from active VNFs and CNFs running on the cloud, display them graphically and save reports for later usage in an easy to read excel and json formats. Supported statistics include Networking, Storage and CPU/MEM.

AIM uses a non-intrusive method, so the cloud users' resources are not at risk as all information is gathered directly from the underlying layers.

AIM is very easy to deploy and remove and very intuitive to use and deploy on your system, the tool's footprint is very small and easily removed using the provided installation script.

Supported platforms:

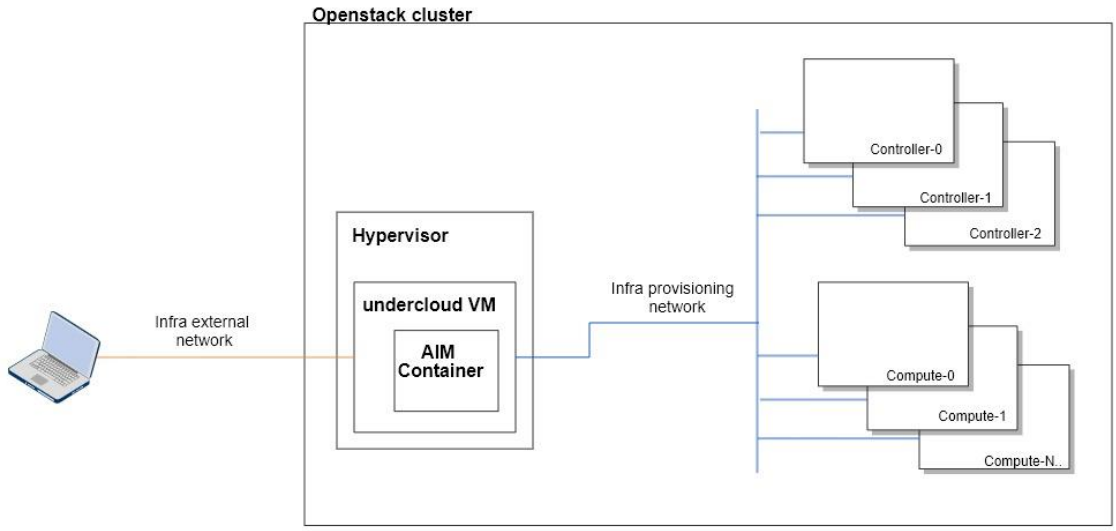
- **CBIS version 19 and above.**
- **NCS version 20 and above.**

1.1. AIM Topology

From topology point of view, AIM uses a simple topology of main backend container which connects to multiple agents distributed on the infrastructure nodes.

A diagram of this topology can be seen below:

AIM topology for Openstack(CBIS)



As displayed in the picture, AIM UI machine resides on the Undercloud.

The agents are containers that run on each of the computes and gather information from the operating system and the virtual resources manager (libvirtd).

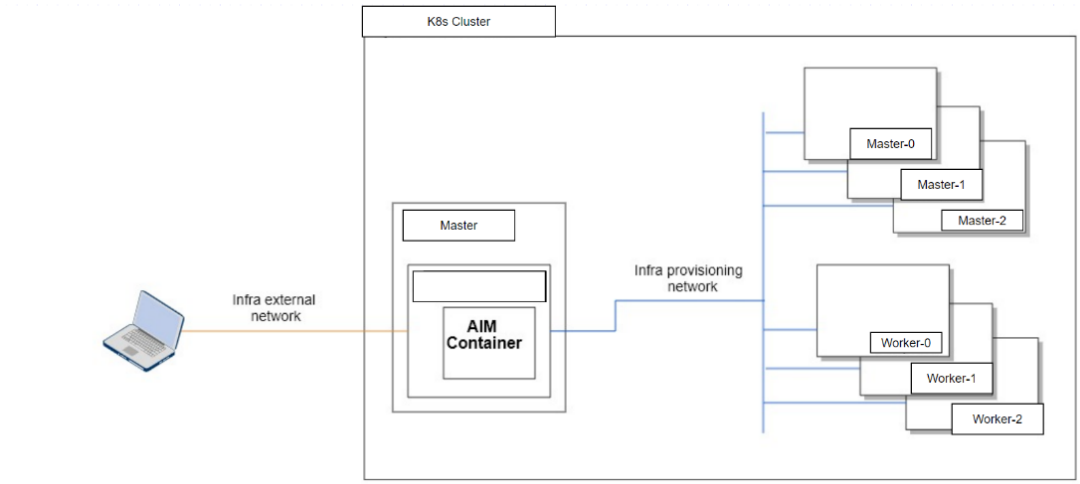
The connection to the agents uses the same ansible structure and parameters used by the Undercloud via the underlay provisioning network.

NOTE:

- AIM UI uses currently hardcoded ports on the Undercloud:
  - `9990` **`main UI`**
  - `13456` **`main backend`**
  - `11234` **`agents`**

- 6543 **RabbitMQ**
- 27017 **mongoDB**
-

- If any of the ports mentioned above **collides** with any of the open ports in the **undercloud,** you can change and overwrite them with your own free ports, more about this in the installation section.



AIM topology for K8S(NCS)

As displayed in the picture, AIM UI/BE machine resides on a container in the master node.

The agents are containers that run on each of the workers/edges/master and gather information from the operating system and the virtual resources manager.

The connection to the agents uses the same ansible structure and parameters used by the kubectl via the underlay provisioning network.

NOTE:

- AIM UI uses currently hardcoded ports on the Worker/Controller:
  - 9990 **main UI**
  - 13456 **main backend**
  - 11234 **agents**
  - 11235 **Stats Collector Server**
  - 6543 **RabbitMQ**
  - 27017 **mongoDB**
- If any of the ports mentioned above **collides** with any of the open ports in the **Master,** you can change and overwrite them with your own free ports, more about this in the installation section.

2. AIM downloads

2.1. Download AIM

The latest AIM version can be downloaded from the link below:

https://artifactory-espoo2.int.net.nokia.com/artifactory/technology-productization-local/aim/production/

2.2. AIM tool binaries

File type	File name
Tool installer script	*aim-install.yml*
Tool package	*aim.tar.gz*
Ansible installer	*ansible_installer*
Ansible libraires	*wheelhouse.tar.gz*

- To verify successful downloads, please verify using the sha files from Artifactory.

3. Installation

3.1 Network requirements

As AIM connects multiple components and has a website there are few requirements from network perspective.

Below is a list of network requirements:

1. Access from your laptop/PC to AIM container website, TCP port 9990
2. Access from AIM container to internal RabbitMQ, TCP port 6543
3. Access from AIM container to internal mongo DB, TCP port 27017
4. Access from AIM container to the local backend, TCP port 13456

The following are default ports used by AIM installation, if any change is needed, please follow below instructions to use any other port rather than the 'static' ports mentioned above:

- When running ansible-playbook aim-install.yml command you can add manually the following arguments in order to change one of the static ports used by aim:

```
1.  ansible-playbook aim-install.yml -e UI_PORT=XXX (Change default UI PORT)
2.  ansible-playbook aim-install.yml -e BE_PORT=XXX (Change default BE PORT)
3. ansible-playbook aim-install.yml -e AGENTS_PORT=XXX (Change default AGENTS
   PORT)
4. ansible-playbook aim-install.yml -e MONGO_PORT=XXX (Change default MONGO PORT)
```

NOTE: In previous versions all the ports where static and couldn't be changed, if a change is needed (in older versions), please contact the AIM team for support as stated in support information section.

3.2 Install procedure

<u>CBIS Section</u>:

To install AIM under CBIS please follow these steps:

1. Check that the ports required are available on your Undercloud VM (see section 3.1)
2. Copy the aim.tar.gz file into the Undercloud server (for ease of use please pre-create a separate folder).
3. Copy the AIM installer script into the Undercloud VM to the same folder as the tar.gz file
4. Run the following command from the same folder where the files reside (see flag reference below):

```
ansible-playbook aim-install.yml -e platform=cbis
```

Script optional arguments:

Flag	Description	mandatory	default
platform	cbis / ncs	no	cbis
action	Install / uninstall	no	install

NCS Section:

To install AIM under NCS please follow these steps:

1. Check that the ports required are available on your Master/Controller (see section 3.1)
2. Copy the aim.tar.gz file into the Master/Controller server (for ease of use please pre-create a separate folder).
3. Copy the AIM installer script into the Master/Controller to the same folder as the tar.gz file
4. Run the following command from the same folder where the files reside (see flag reference below):

```
ansible-playbook aim-install.yml -e platform=ncs
```

NCS credentials

- AIM uses ssh private key to connect to agents by default, it uses the default private key in the **~/.ssh/ folder**
- If you use different key than the one in ~/.ssh/ folder, please use the following argument in the install command: **-e ssh_private_key=/your/path/private_key.pem**
- The default username AIM use is the one you are logged in from, to change that please use the following argument: **-e username=<username>**
- If you don't use ssh keys, and would like to use username and password, please use the following arguments: **-e username=<username> -e password=<password>**
- If you don't want to enter the password to command line, please use this argument: **-e set_creds=true** , this argument will give a prompt to enter username and password
- so to summarize, to use these arguments you add them to this command:
    a. `ansible-playbook aim-install.yml -e platform=ncs <ARGUEMNTS>`

Script optional arguments:

Flag	Description	mandatory	default
platform	cbis / ncs	no	Cbis
action	Install / uninstall	no	Install

CNA Section:

To install AIM under CNA please follow these steps:

- Check that the ports required are available on your Master/Controller (see section 3.1)
- Copy the `aim.tar.gz` file into the Master/Controller server (for ease of use please pre-create a separate folder).
- Copy the `AIM installer` script into the Master/Controller to the same folder as the tar.gz file
- Copy the `Ansible installer` script into the Master/Controller to the same folder as the tar.gz file
- Copy the `Ansible libraires` script into the Master/Controller to the same folder as the tar.gz file
- Run the following commands from the same folder where the files reside (see flag reference below):
    a. source `ansible_installer`
    b. `ansible-playbook aim-install.yml -e platform=ncs`
    c. please read the ncs credentials section above
    d. `source ansible_installer uninstall`

aim Installer script optional arguments:

Flag	Description	mandatory	default
platform	cbis / ncs	no	cbis
action	Install / uninstall	no	install

After installation you will get a UI URL, please use this URL to access the UI (check AIM usage section 4)

```
"AIM is available at http://10.4.165.66:9099"
```

3.3 Uninstall procedure

To uninstall AIM on CBIS or NCS (not CN-A) please follow these steps:

1. Run the following command from the same folder where the files reside (see flag reference below):

   ```
   ansible-playbook aim-install.yml -e action=uninstall -e platform=cbis/ncs
   ```

To uninstall on CN-A please follow these steps:

1. Run the following command from the same folder where the files reside (see flag reference below):
   a. `source ansible_installer`
   b. `ansible-playbook aim-install.yml -e action=uninstall -e platform=ncs`
   c. `source ansible_installer uninstall`

   Script optional arguments:

Flag	Description	mandatory	default
platform	cbis / ncs	no	cbis
action	Install / uninstall	no	install

3.4 Memory limitation

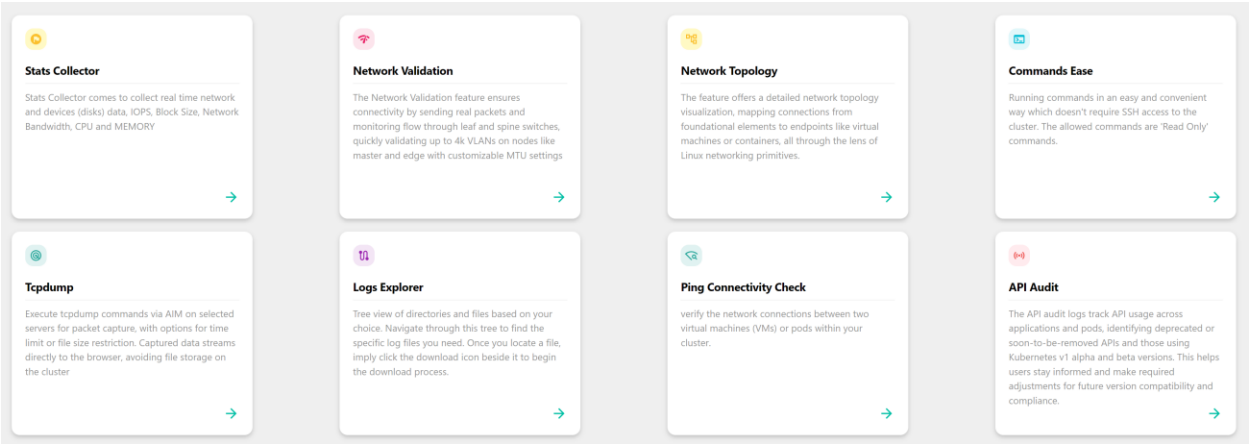
We limit the RAM memory for the aim backend container, this affects how much mongo can store data in memory, the default values are (which are recommend):

ALL Memory	Mongo RAM	Other container services RAM
16GB	12GB	4GB

This value can be changed via installation:

```
ansible-playbook aim-install.yml -e platform=ncs/cbis -e memory_limit=<int
```

4. AIM usage and features



AIM has multiple features:

1- Stats Collector
1.1- Sample Statistic (ncs only)
2- Network Validation
3- Network Topology
4- Command Ease (ncs only)
5- Tcpdump
6- Log Explorer (ncs only)
7- Ping Connectivity Check
8- API Audit (ncs only)

4.1. Network topology

The first functionality of AIM is to display the environment topology both from "infrastructure" perspective in which the tool displays the infrastructure resources (such as computes, controllers, and nodes) and "user" perspective in which the user's resources are displayed (VMs, VNFs, PODs etc.).

Once the user opens the tools web site that's the first screen displayed.

Below you can see the topology screen example:

As can be seen the screen is split into a few sections

- **Environment structure tree** – (top left side) in which a list of the infrastructure resources is displayed and allow the user selecting specific resources to view.
  In this section the whole infrastructure resources list is displayed, it allows the user, expending, collapsing, and selecting specific resources (currently – computes).
- **Resource network topology** – (right side) once the user clicked on a specific resource from the tree, its network topology is displayed. It allows to the user to view the whole structure of network inside the target compute including the OVS-switch entities and interfaces connected to them up to VMs that reside on the target compute. In addition, the user can zoom in and out using the +/- on the bottom left side, centering the view and locking the editing of the view.
- **Resource detail view** – (bottom left side) once the user clicks on a specific resource from the tree view its details will be shown here. Current displayed details are UUID, Tenant, status, power state
- **Network topology mini-map** – (bottom right side) This display allows the user to see the whole topology view his main view relative location in it.

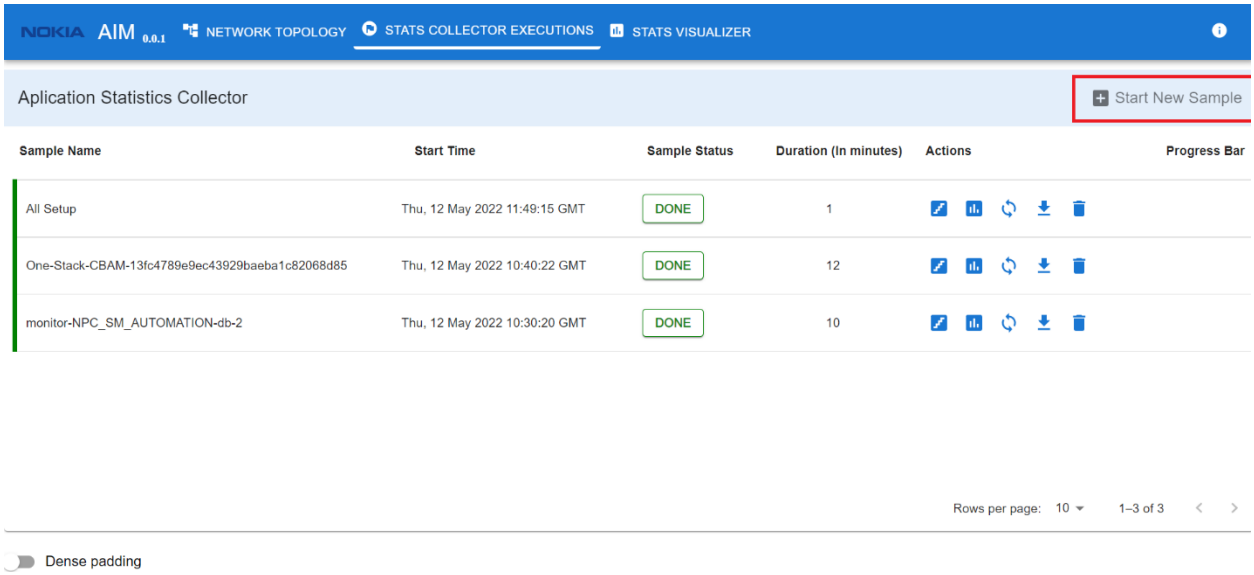
4.2. Application Stats Collector

In addition to displaying the environment topology the user might want to gather statistics from the user resources. This is done by clicking on the "Stats Collector Executions" button at the top menu.

From here the user can manage his stats collecting executions.

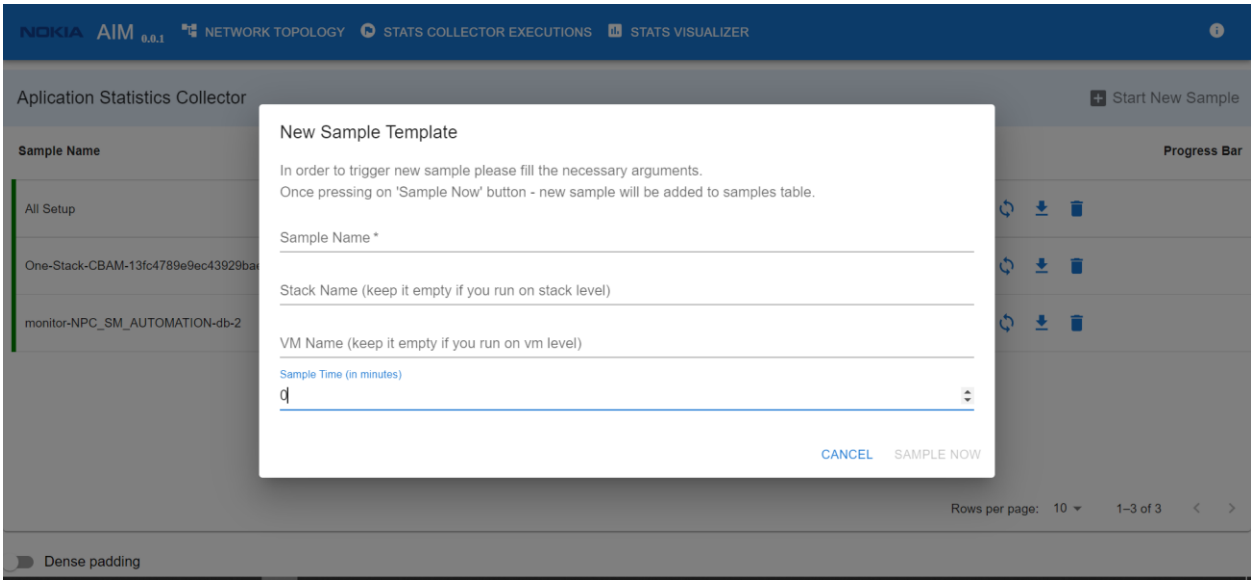
4.2.1. Running execution

To run stats collection, click on the "start new sample" button on the right top of the screen, as seen below:



This will open the execution definition screen where the user can define the desired collection

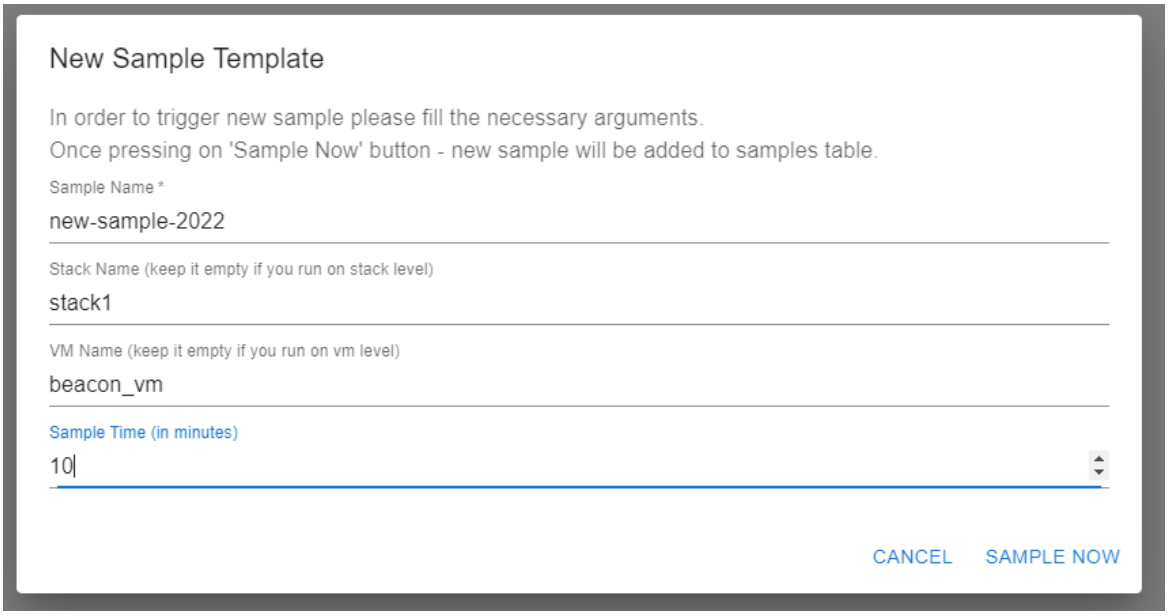
See example below:



The details needed for execution are:

- **Sample name** (mandatory) – name your execution
- **Stack name** (optional) – fill this if the desired resource is a stack (or stacks).
- **VM name** (optional) – fill this if the desired resource is a VM (or VMs).
- **Sample time** (mandatory) – fille this to specify the time duration of the sample in minutes.

See screenshot below for example:



NOTE: when executing a sample at least one of the fields stack name or vm name need to be filled.

NOTE: In case user want to sample multiple resources (Stacks or VMs) it can be achieved using the below regular expression options:

- **p: - use this for setting prefix, for example p:vm_ will select all VMs with name that start with vm_ , the same goes for "Stack name" field.**
- **s: - use this for setting suffix, for example s:_1 will select all VMs with name that start with _1 , the same goes for "Stack name" field.**
- **r: - use this for using regex, following common regex specifications.**

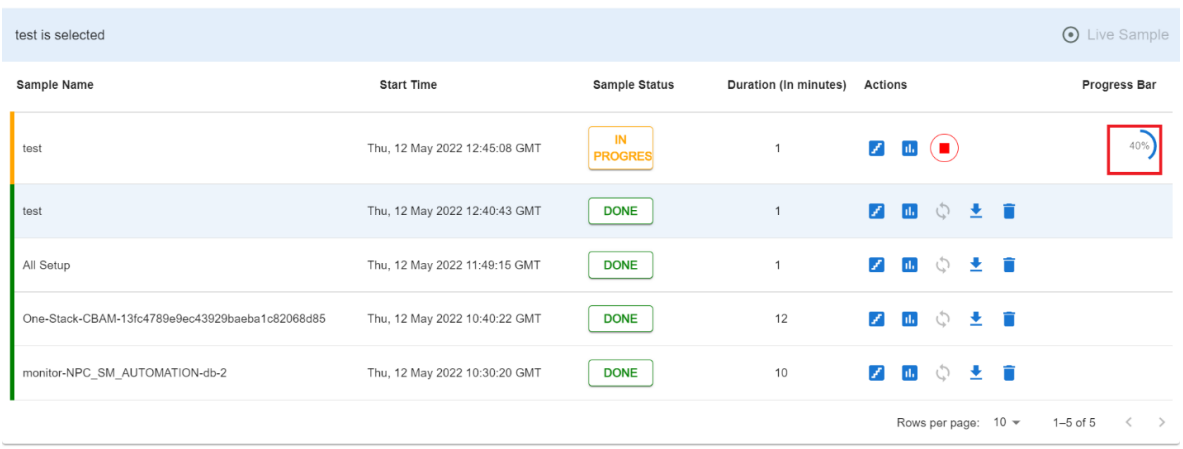
After filling all required details, click on "SAMPLE NOW" to start the sample.

NOTE: Only one sample can be active at any time.

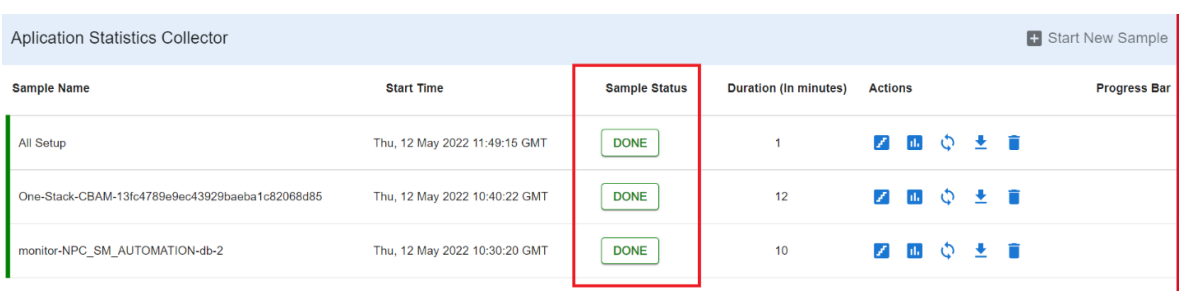
Once the execution started the user is displayed with a list of the pre-requisites steps which are done by the tool, after all steps are finished the user is moved automatically to the sample list view after a few seconds.

In the list view the user can see a few quick details such as:

- sample progress (see below)

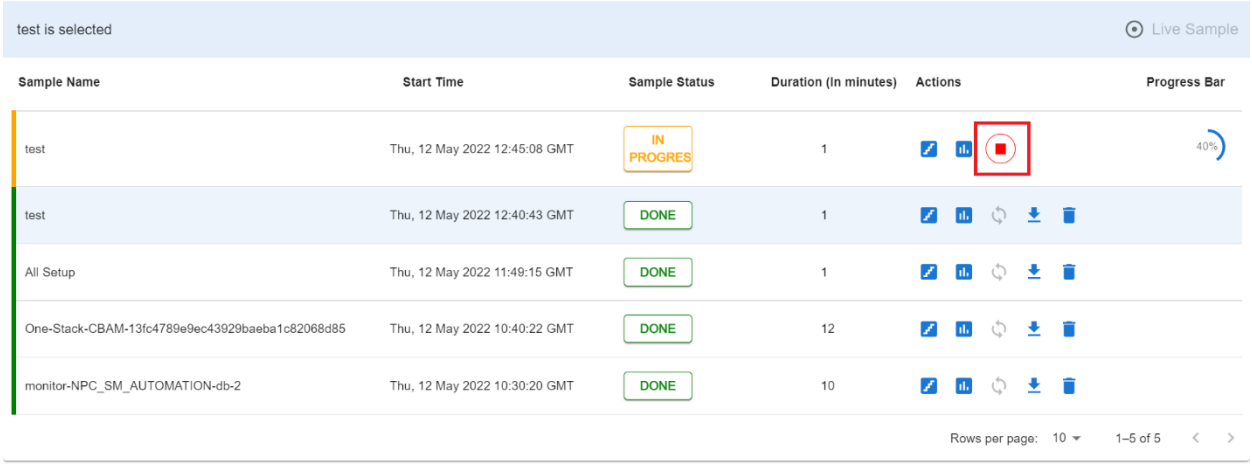


- Sample status (see below)



4.2.2. Stopping execution

Once a sample started it's possible to stop it by using the button displayed below.



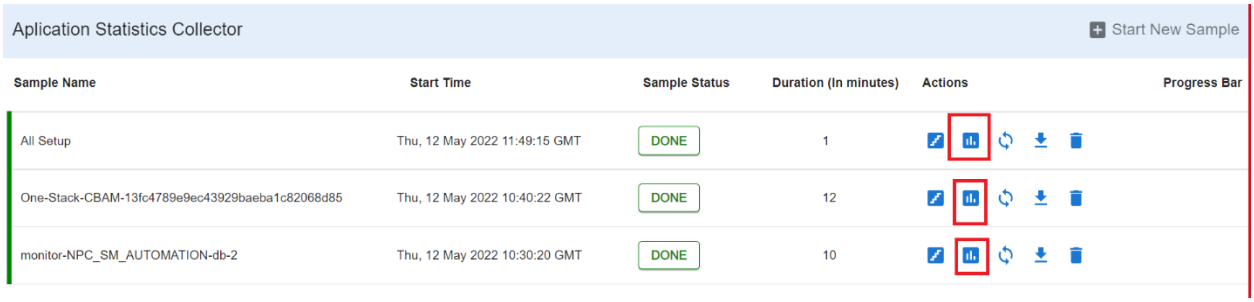
NOTE: Stopped sample isn't deleted but is displayed in the sample list view.

4.2.3. Displaying live statistics

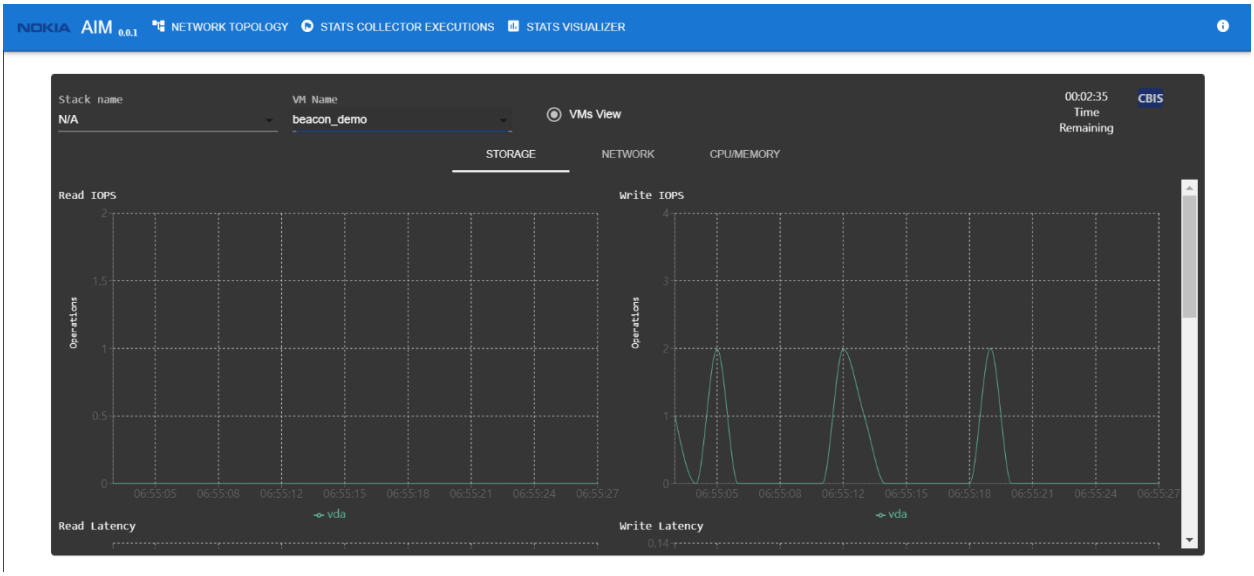
Once a sample started the user can display its current stats in the visualizer screen.

the user will be moved automatically to eh sample graphic view.

To manually move to the visualizer screen, click on the "Graph" button on the sample card as displayed below

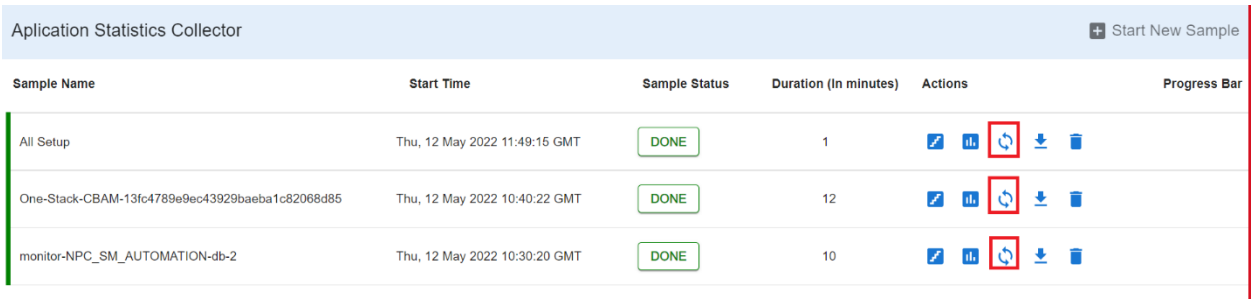


An Example of the sample visualizer screen can be seen below



4.2.4. Rerun sample

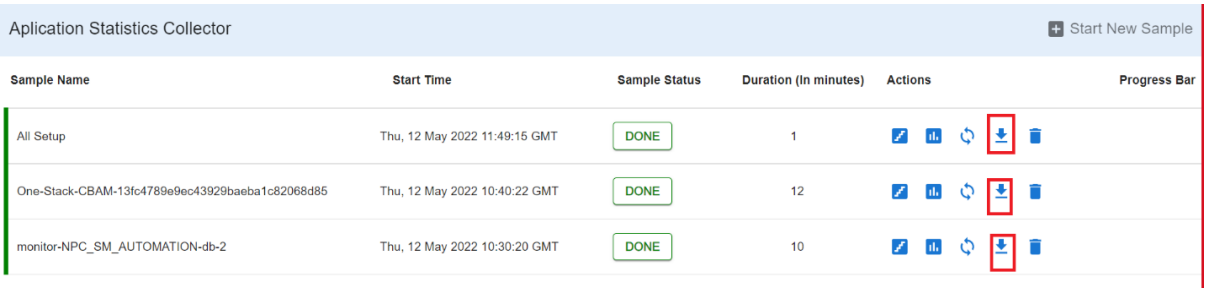
The user is able to re-execute a past sample with one click. To re-run a sample click on the "Re-run" button on the sample card as displayed below.



4.2.5. Download result file

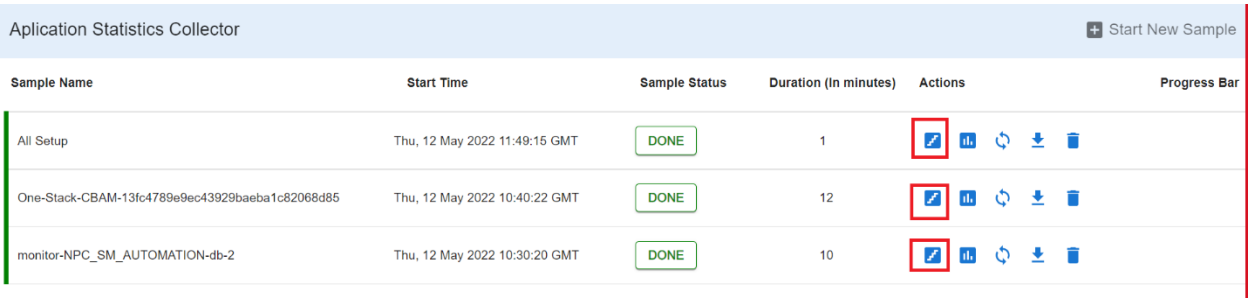
Once a sample started the user can download the results of the execution.

To download results, click on the "Download" button on the sample card as displayed below.



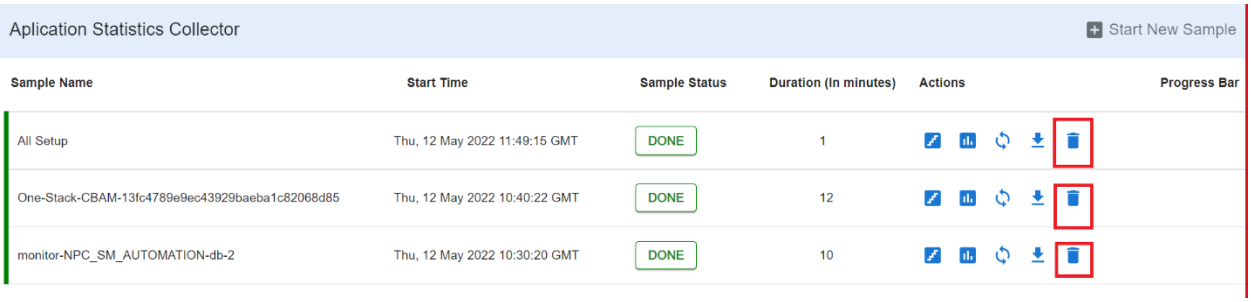
4.2.6. Checking sample steps

Once a sample has started sometimes there are failures. To debug the sample steps and check why and where it failed along the way please click on the "Sample steps" icon as displayed below



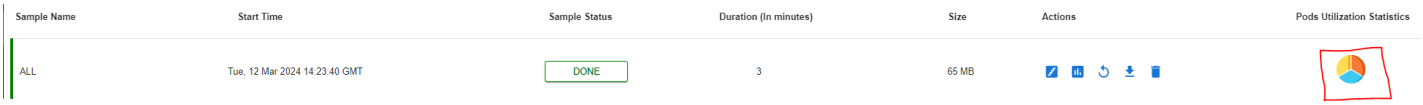
4.2.7. Deleting a sample

Sometimes the user wants to delete samples for any reason. To delete a sample, click on the "Delete" icon on the sample card as displayed below.



4.2.8. Sample Statistics

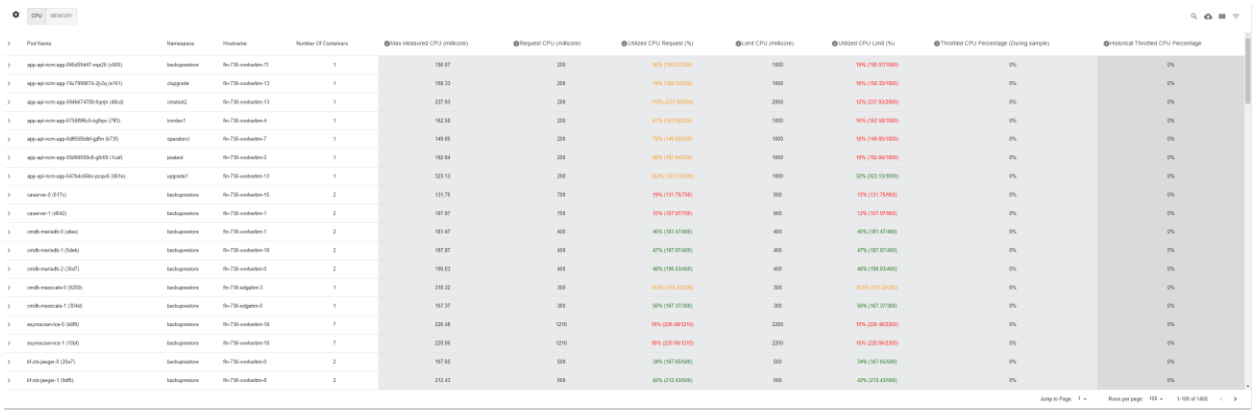
This feature is NCS only



Sample Name	Start Time	Sample Status	Duration (In minutes)	Size	Actions	Pods Utilization Statistics
ALL	Tue, 12 Mar 2024 14:23:40 GMT	DONE	3	65 MB		

After each sample is Done or stopped, you can press on the pod Utilization statistics and get a wholesome view of Pods statistics during the sample.

CPU



Pod Name	Namespace	Hostname	Number Of Containers	Max Measured CPU (millicore)	Request CPU (millicore)	Utilized CPU Request (%)	Limit CPU (millicore)	Utilized CPU Limit (%)	Throttled CPU Percentage (During sample)	Historical Throttled CPU Percentage
app-api-ncm-app-646d59d47-nqx26 (c568)	backuprestore	fln-736-workerbm-11	1	190.87	200	95% (190.87/200)	1000	19% (190.87/1000)	0%	0%
app-api-ncm-app-7dc7999674-2jr2q (e161)	clupgrade	fln-736-workerbm-13	1	158.33	200	79% (158.33/200)	1000	16% (158.33/1000)	0%	0%
app-api-ncm-app-594667478f-6qnjn (d8cd)	cmatest2	fln-736-workerbm-13	1	237.93	200	119% (237.93/200)	2000	12% (237.93/2000)	0%	0%
app-api-ncm-app-67569f6c5-kg8qn (7ff3)	lcmder1	fln-736-workerbm-4	1	162.58	200	81% (162.58/200)	1000	16% (162.58/1000)	0%	0%
app-api-ncm-app-6d9550d8d-jgfhn (b739)	operatorcl	fln-736-workerbm-7	1	149.85	200	75% (149.85/200)	1000	15% (149.85/1000)	0%	0%
app-api-ncm-app-59d9955fcd-g6r68 (1caf)	paatest	fln-736-workerbm-2	1	192.84	200	96% (192.84/200)	1000	19% (192.84/1000)	0%	0%
app-api-ncm-app-647b4d98bc-pcqv6 (8f1e)	upgrade1	fln-736-workerbm-13	1	323.13	200	162% (323.13/200)	1000	32% (323.13/1000)	0%	0%
casemer-0 (b17c)	backuprestore	fln-736-workerbm-15	2	131.75	700	19% (131.75/700)	500	15% (131.75/500)	0%	0%
casemer-1 (d842)	backuprestore	fln-736-workerbm-1	2	197.97	700	15% (197.97/700)	500	12% (197.97/500)	0%	0%
cmdb-mariadb-0 (afea)	backuprestore	fln-736-workerbm-1	2	181.47	400	45% (181.47/400)	400	45% (181.47/400)	0%	0%
cmdb-mariadb-1 (5de4)	backuprestore	fln-736-workerbm-18	2	187.87	400	47% (187.87/400)	400	47% (187.87/400)	0%	0%
cmdb-mariadb-2-(35d7)	backuprestore	fln-736-workerbm-0	2	190.03	400	48% (190.03/400)	400	48% (190.03/400)	0%	0%
cmdb-mariscale-0 (5209)	backuprestore	fln-736-edgebm-3	1	319.32	300	103% (319.32/300)	300	103% (319.32/300)	0%	0%
cmdb-mariscale-1 (3044)	backuprestore	fln-736-edgebm-0	1	167.37	300	56% (167.37/300)	300	56% (167.37/300)	0%	0%
esymaceservice-0 (b8f9)	backuprestore	fln-736-workerbm-18	7	226.48	1210	19% (226.48/1210)	2200	10% (226.48/2200)	0%	0%
esymaceservice-1 (10fd)	backuprestore	fln-736-workerbm-18	7	220.95	1210	18% (220.95/1210)	2200	10% (220.95/2200)	0%	0%
kf-cts-jaeger-0 (20e7)	backuprestore	fln-736-workerbm-0	2	167.65	500	34% (167.65/500)	500	34% (167.65/500)	0%	0%
kf-cts-jaeger-1 (5df5)	backuprestore	fln-736-workerbm-8	2	212.43	500	42% (212.43/500)	500	42% (212.43/500)	0%	0%

Here it shows all the pods that you requested to sample, for each pod it shows metrics related to CPU.

- Number of containers
  - Number of containers the pod has

- Max Measured CPU (millicore)
  - The maximum cpu utlilization that has been measured for the pod during the sample, in millicore

- Request CPU (millicore)
  - The request CPU that has been defined for the pod during creation.

- Utilized CPU Request (%)
  - The utilization percentage of the max measured CPU from the request CPU

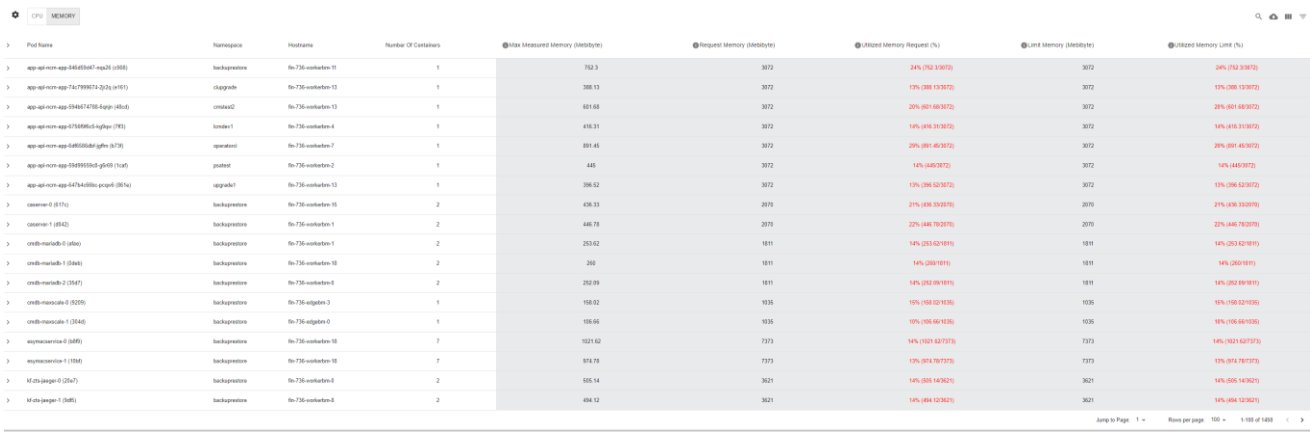
- Limit CPU (millicore)
  - The limit CPU that has been defined for the pod during creation.

- Utilized CPU limit (%)
    - The utilization percentage of the max measured CPU from the limit CPU

- Throttled CPU Percentage (%) during sample
    - This metric gauges the proportion of a pod/container was throttled during a given observation window (the sample duration), offering a clear view of how resource limitations might be affecting your container or pod's efficiency, calculated by throttled period divided by runnable period.

- Historical Throttled CPU Percentage (%)
    - This metric gauges the proportion of a pod/container was throttled over a historical period (since pod/container creation), offering a clear view of how resource limitations might be affecting your container or pod's efficiency, calculated by throttled period divided by runnable period.

Memory



CPU MEMORY

Pod Name	Namespace	Hostname	Number Of Containers	Max Measured Memory (Mebibyte)	Request Memory (Mebibyte)	Utilized Memory Request (%)	Limit Memory (Mebibyte)	Utilized Memory Limit (%)
app-api-ncm-app-846459d47-wqx26 (c958)	backuprestore	fin-736-workerbm-11	1	752.3	3072	24% (752.3/3072)	3072	24% (752.3/3072)
app-api-ncm-app-74c799967-2jr2q (e161)	clupgrade	fin-736-workerbm-13	1	388.13	3072	13% (388.13/3072)	3072	13% (388.13/3072)
app-api-ncm-app-594867478-6qnjn (48cd)	cmstest2	fin-736-workerbm-13	1	601.68	3072	20% (601.68/3072)	3072	20% (601.68/3072)
app-api-ncm-app-67566ff6c5-kgfqv (7ff3)	lcmdev1	fin-736-workerbm-4	1	416.31	3072	14% (416.31/3072)	3072	14% (416.31/3072)
app-api-ncm-app-6df6508dbf-jgffm (b73f)	operator2	fin-736-workerbm-7	1	891.45	3072	29% (891.45/3072)	3072	29% (891.45/3072)
app-api-ncm-app-5fd99515c8-p5r59 (1caf)	pcatest	fin-736-workerbm-2	1	445	3072	14% (445/3072)	3072	14% (445/3072)
app-api-ncm-app-647b4c988c-pcqn6 (861e)	upgrade1	fin-736-workerbm-13	1	396.52	3072	13% (396.52/3072)	3072	13% (396.52/3072)
caserver-0 (617c)	backuprestore	fin-736-workerbm-16	2	436.33	2070	21% (436.33/2070)	2070	21% (436.33/2070)
caserver-1 (d842)	backuprestore	fin-736-workerbm-1	2	446.78	2070	22% (446.78/2070)	2070	22% (446.78/2070)
cmdb-mariadb-0 (a9ab)	backuprestore	fin-736-workerbm-1	2	253.62	1811	14% (253.62/1811)	1811	14% (253.62/1811)
cmdb-mariadb-1 (0deb)	backuprestore	fin-736-workerbm-18	2	260	1811	14% (260/1811)	1811	14% (260/1811)
cmdb-mariadb-2 (35d7)	backuprestore	fin-736-workerbm-8	2	252.09	1811	14% (252.09/1811)	1811	14% (252.09/1811)
cmdb-mariscale-0 (9259)	backuprestore	fin-736-edgebm-3	1	158.02	1035	15% (158.02/1035)	1035	15% (158.02/1035)
cmdb-mariscale-1 (304d)	backuprestore	fin-736-edgebm-0	1	106.66	1035	10% (106.66/1035)	1035	10% (106.66/1035)
esymaceservice-0 (bdf9)	backuprestore	fin-736-workerbm-18	7	1021.62	7373	14% (1021.62/7373)	7373	14% (1021.62/7373)
esymaceservice-1 (1f8d)	backuprestore	fin-736-workerbm-18	7	974.78	7373	13% (974.78/7373)	7373	13% (974.78/7373)
kf-cts-jaeger-0 (2fa7)	backuprestore	fin-736-workerbm-8	2	505.14	3621	14% (505.14/3621)	3621	14% (505.14/3621)
kf-cts-jaeger-1 (9df6)	backuprestore	fin-736-workerbm-8	2	494.12	3621	14% (494.12/3621)	3621	14% (494.12/3621)

Jump to Page: 1 Rows per page: 100 1-100 of 1458

Here it shows all the pods that you requested to sample, for each pod it shows metrics related to Memory.

- Number of containers
    - Number of containers the pod has

- Max Measured memory (Mebibyte)
    - The maximum memory utlilization that has been measured for the pod during the sample, in Mebibyte.

- Request Memory (Mebibyte)
    - The request Memory that has been defined for the pod during creation.

- Utilized Memory Request (%)
    - The utilization percentage of the max measured Memory from the request Memory

- Limit Memory (Mebibyte)
    - The limit Memory that has been defined for the pod during creation.

- Utilized Memory Limit (%)
    - The utilization percentage of the max measured Memory from the Limit Memory

If the pod has more than one container, you can check each container metrics:

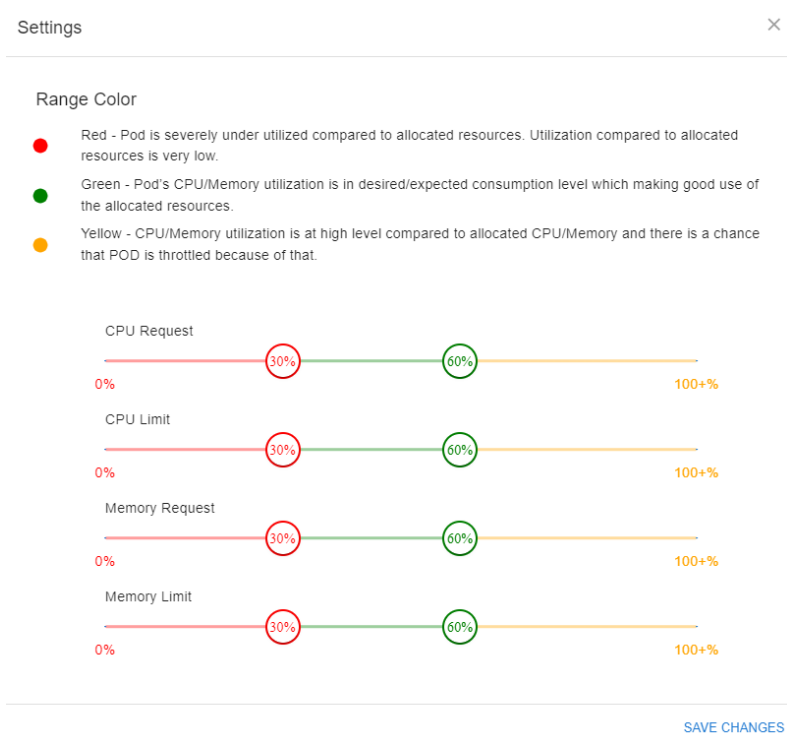
Pod Name	Namespace	Hostname	Number Of Containers	Max Measured CPU (millicore)	Request CPU (millicore)	Utilized CPU Request (%)	Limit CPU (millicore)	Utilized CPU Limit (%)	Throttled CPU Percentage (During sample)	Historical Throttled CPU Percentage
zts-webserver-1 (1task)	maintenance1	fin-736-worker6n-9	6	99.62	330	30% (99.62/330)	600	16% (99.62/600)	0%	0%
chora-sidecar				1.21	20	6% (1.21/20)	20	6% (1.21/20)	0%	0%
registrationcontainer				0.11	10	1% (0.11/10)	10	1% (0.11/10)	0%	0%
webserver				4.8	200	2% (4.8/200)	500	1% (4.8/500)	0%	0%
fluent-bit				17.98	25	72% (17.98/25)	50	36% (17.98/50)	0%	0%
consulagent-container				46.55	25	186% (46.55/25)	50	93% (46.55/50)	0%	0%
envoy-sidecar				52.92	50	106% (52.92/50)	50	106% (52.92/50)	0%	0%

Pod Name	Namespace	Hostname	Number Of Containers	Max Measured Memory (Mebibyte)	Request Memory (Mebibyte)	Utilized Memory Request (%)	Limit Memory (Mebibyte)	Utilized Memory Limit (%)
integration-6f9ff7b549-29679 (acc7)	maintenance1	fin-736-worker6n-5	9	493.67	2782	18% (493.67/2782)	3082	16% (493.67/3082)
zmsettings				89.8	282	32% (89.8/282)	282	32% (89.8/282)
fluentdconfig				31.13	200	16% (31.13/200)	500	6% (31.13/500)
consulagent-container				62.1	282	22% (62.1/282)	282	22% (62.1/282)
dynamicadaptation				38.69	282	14% (38.69/282)	282	14% (38.69/282)
fluent-bit				38.74	564	7% (38.74/564)	564	7% (38.74/564)
envoy-sidecar				189.46	564	34% (189.46/564)	564	34% (189.46/564)

Its shows all the containers' names and the same kind of metrics we showed for the pod

This is important since some times you can see that a container is throttled but the pod isn't.

for the Utilized CPU Request (%) and Utilized CPU limit (%), we show colors that indicates the severity of the situation, also in settings you can configure the range for these colors:



To show data of containers, when making new sample please check the container level box.

4.2.9. Stats Visualizer

AIM includes the functionality like the classic application stats collector visualizer where the user can upload result files from past executions and load them into the tool to display them graphically.

This function can help users that have old, executed sample files from customer sites but don't have the AIM tool itself available for them, that way they can take their saved samples and upload them to the visualizer to display them graphically.

NOTE: The AIM visualizer file format is tar.gz

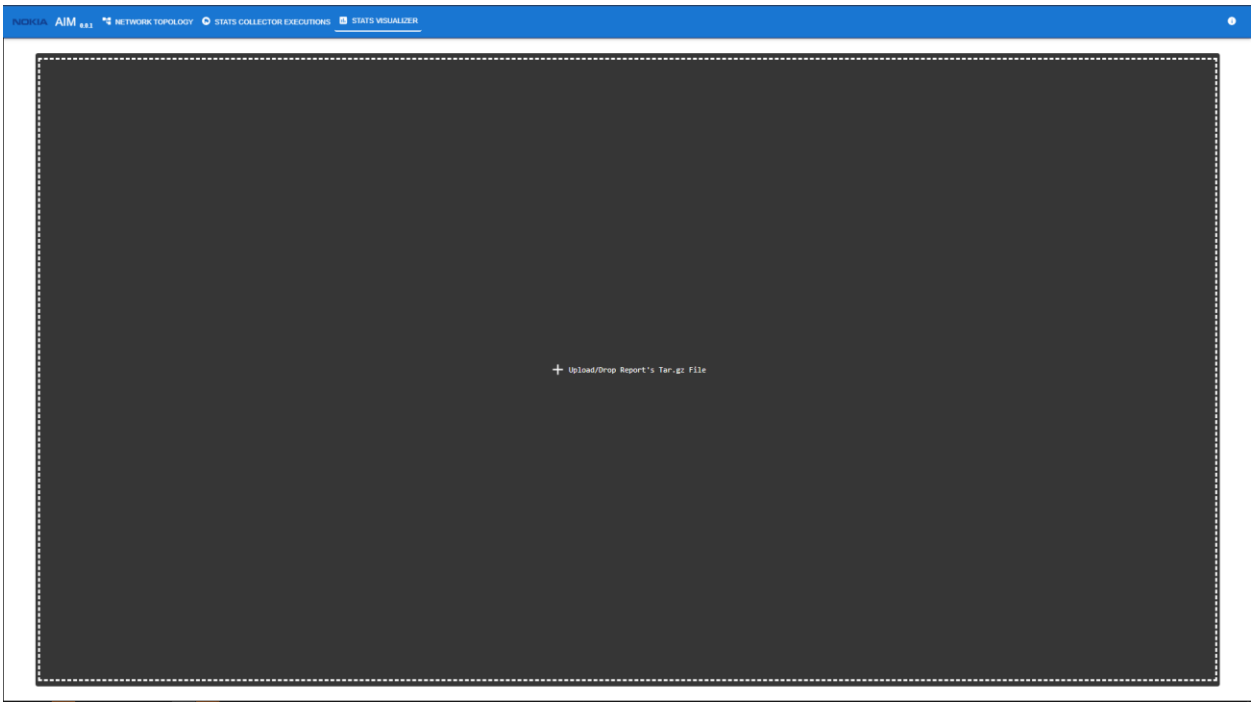
To open the visualizer screen, click on the "STATS Visualizer" button at the top menu.

to upload a file either click on the upload file button and select the file you want or, drag and drop the file to the visualizer screen.

Once the user uploaded a file, AIM parses it and inserts a new entry to the "STATS COLLECTOR EXECUTIONS" with the name "Temporary sample (uploaded manually)"

NOTE: as the system allows one upload at time every consequent upload will overwrite the last uploaded one.

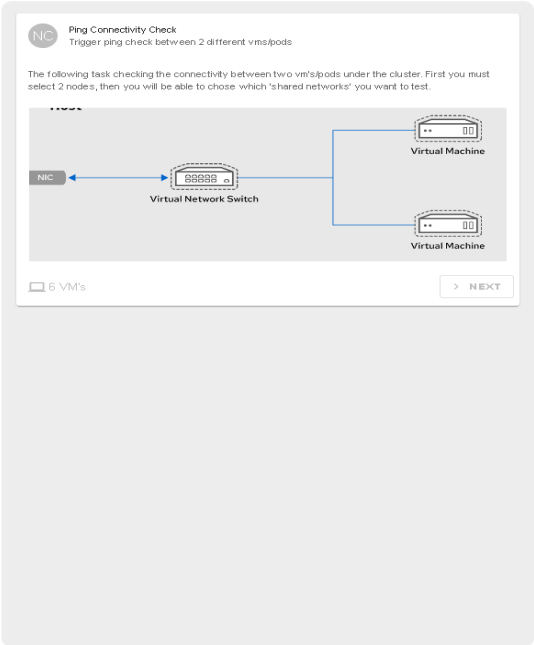
See example below

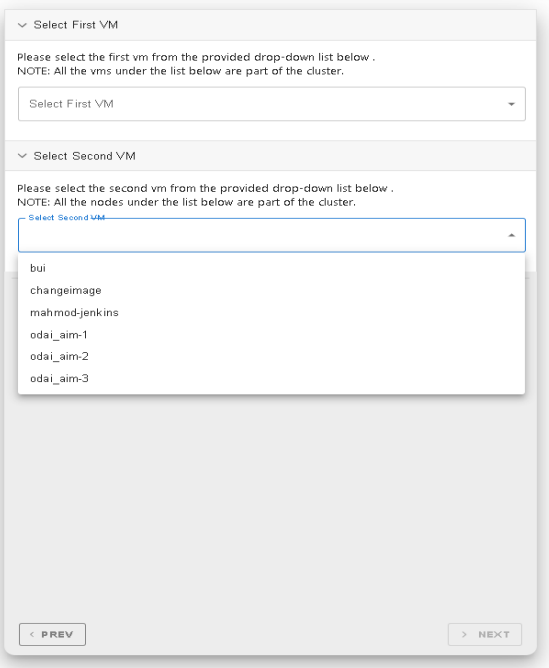


Network Connectivity

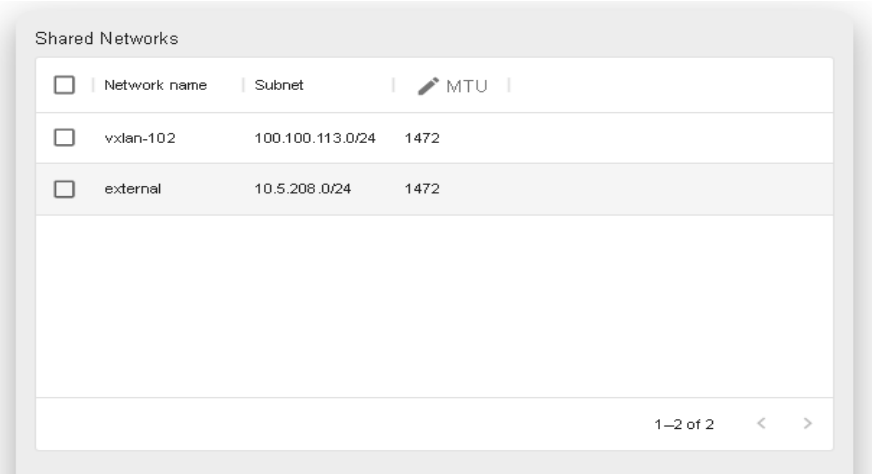
AIM provide basic network connectivity checks functionality between two different nodes (vms / pods) depend on the platform (CBIS / NCS).

First the user must select two different nodes from the whole cluster.





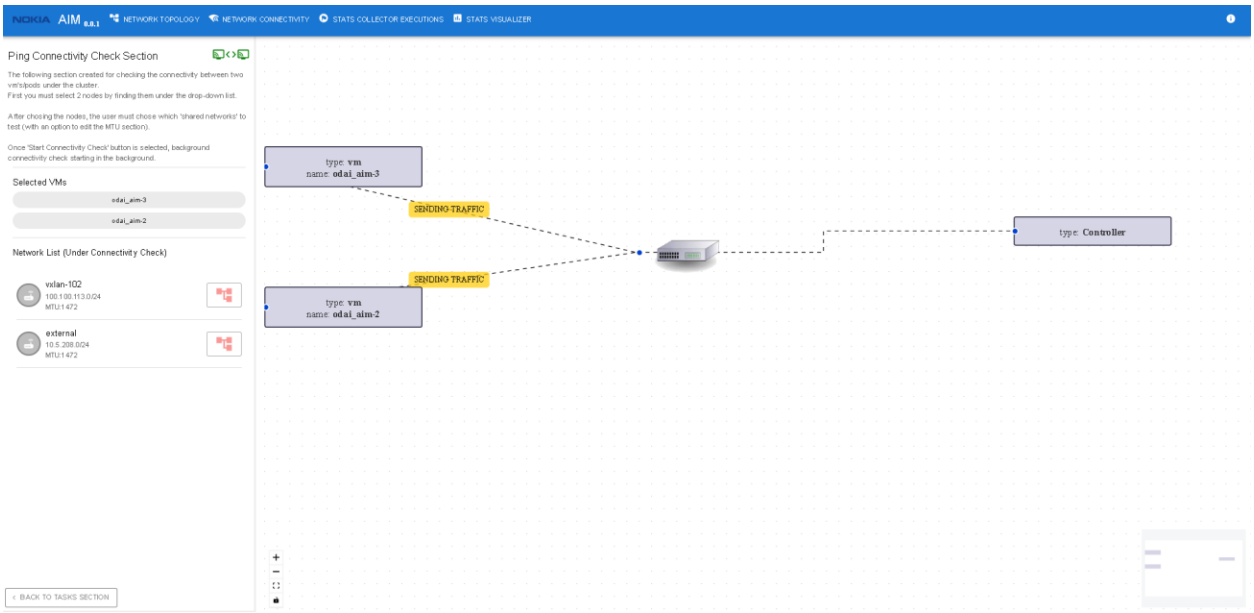
Second, AIM presenting all the 'shared networks' between the **two selected nodes**.



Now when all the 'shared networks' are presented under the GUI, the user is able to select the dedicated networks in order to start the ping test on them.

Another option is to change the MTU size (per each separate network), the MTU size parameter will be used once we trigger the ping test for each different subnet.

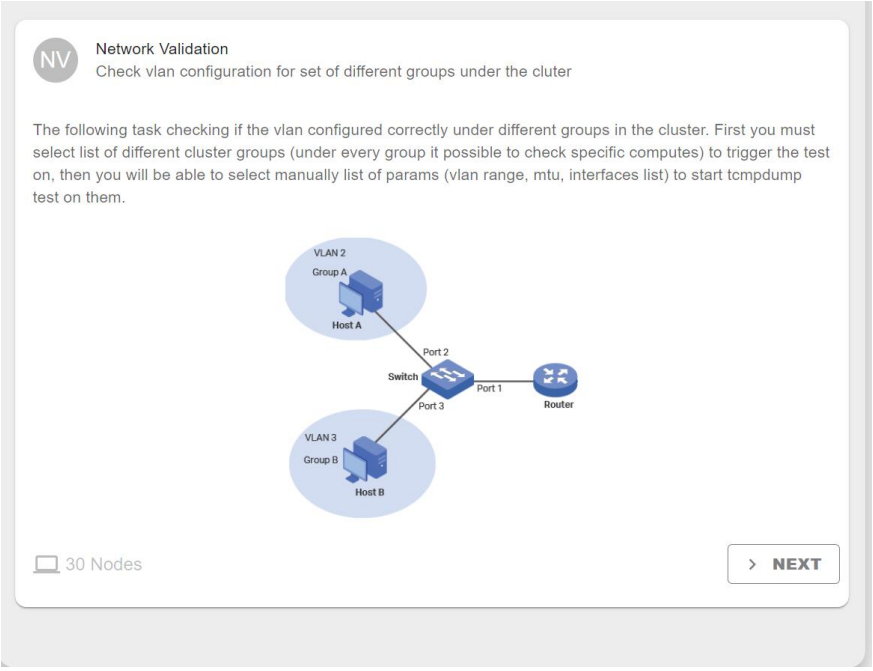
Once the ping test ends, AIM presenting the results of each test by graph visualization.

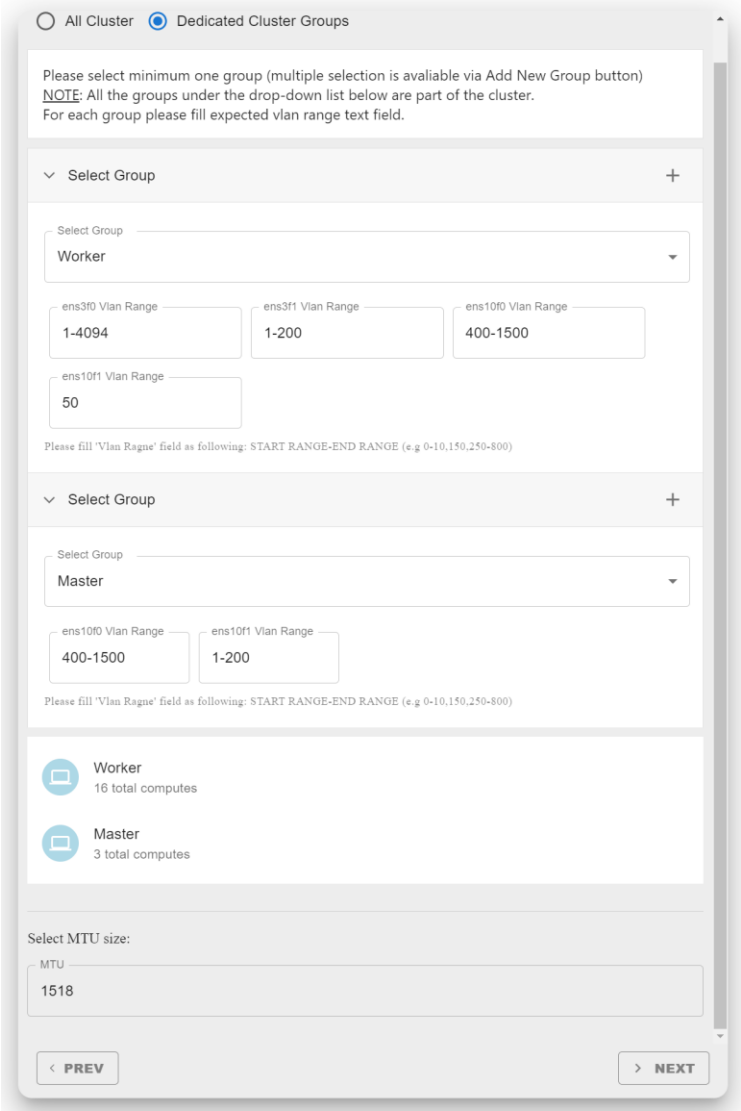


4.4 Network Validation

AIM provides network validation feature, using this feature you can do the checking if the vlan configured correctly under different groups in the cluster. First you must select list of

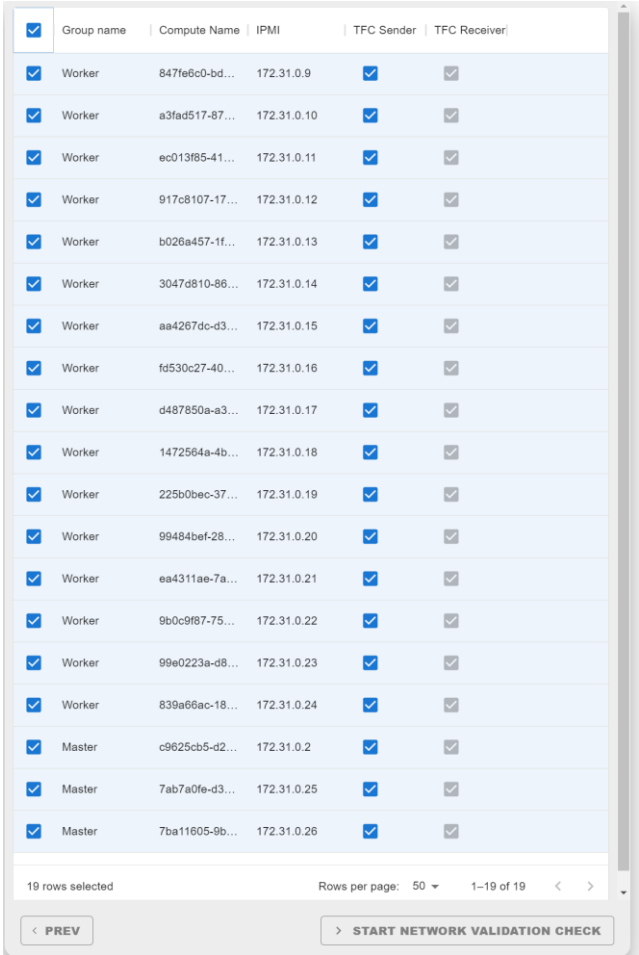
different cluster groups (under every group it possible to check specific nodes) to trigger the test on, then you will be able to select manually list of params (vlan range, mtu, interfaces list) to start the test on them.





Here we choose what groups we want to trigger, each group has different set of interfaces, so we choose what are the expected vlan ranges, so we can compare with results if we got what we expected to be open.

We also choose the MTU size we want to send the L2 packets, so we can also validate the MTU size on the interface



Group name	Compute Name	IPMI	TFC Sender	TFC Receiver
Worker	847fe6c0-bd...	172.31.0.9	✓	✓
Worker	a3fad517-87...	172.31.0.10	✓	✓
Worker	ec013f85-41...	172.31.0.11	✓	✓
Worker	917c8107-17...	172.31.0.12	✓	✓
Worker	b026a457-1f...	172.31.0.13	✓	✓
Worker	3047d810-86...	172.31.0.14	✓	✓
Worker	aa4267dc-d3...	172.31.0.15	✓	✓
Worker	fd530c27-40...	172.31.0.16	✓	✓
Worker	d487850a-a3...	172.31.0.17	✓	✓
Worker	1472564a-4b...	172.31.0.18	✓	✓
Worker	225b0bec-37...	172.31.0.19	✓	✓
Worker	99484bef-28...	172.31.0.20	✓	✓
Worker	ea4311ae-7a...	172.31.0.21	✓	✓
Worker	9b0c9f87-75...	172.31.0.22	✓	✓
Worker	99e0223a-d8...	172.31.0.23	✓	✓
Worker	839a66ac-18...	172.31.0.24	✓	✓
Master	c9625cb5-d2...	172.31.0.2	✓	✓
Master	7ab7a0fe-d3...	172.31.0.25	✓	✓
Master	7ba11605-9b...	172.31.0.26	✓	✓

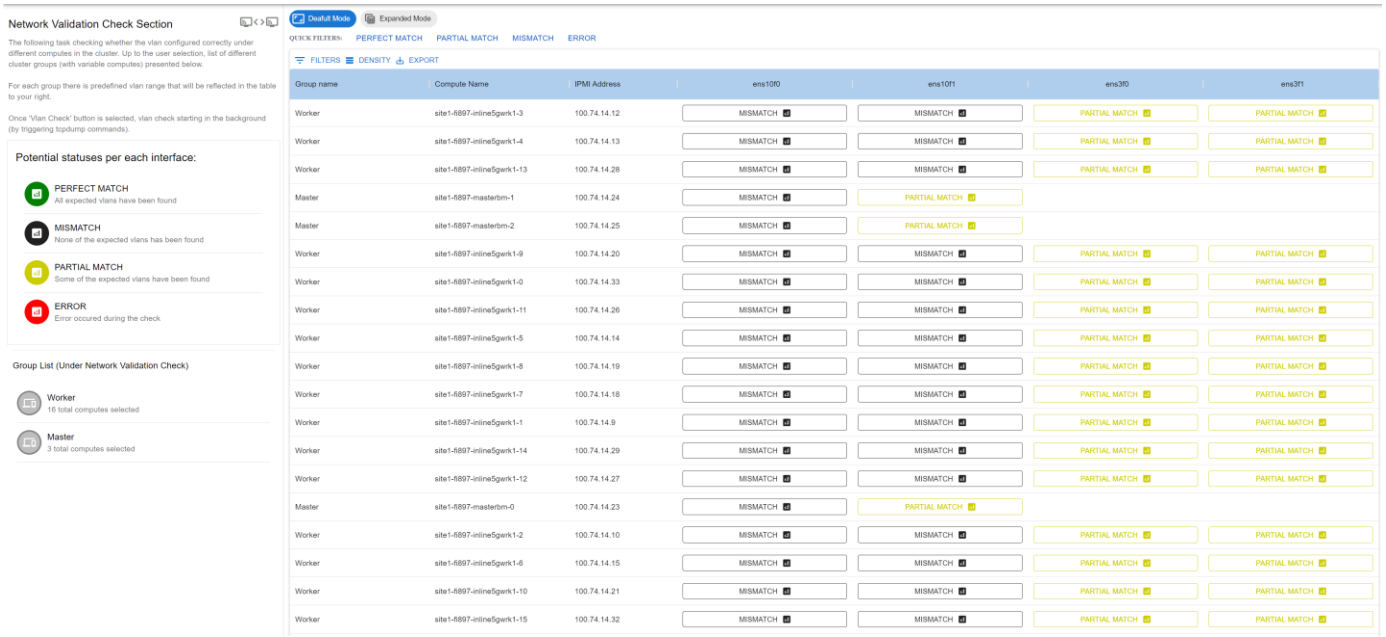
19 rows selected Rows per page: 50 1–19 of 19

PREV START NETWORK VALIDATION CHECK

Here we choose what servers from these groups we want to check, we also choose which one of them we want to send packets from(default all sends), this way by choosing source we can validate multi rack validation in switch(vlans open/closed).

Then we start the test:

We wait till we get the results:



The result can show us for each server we chose, for each interface, if we got the expected VLAN or not:

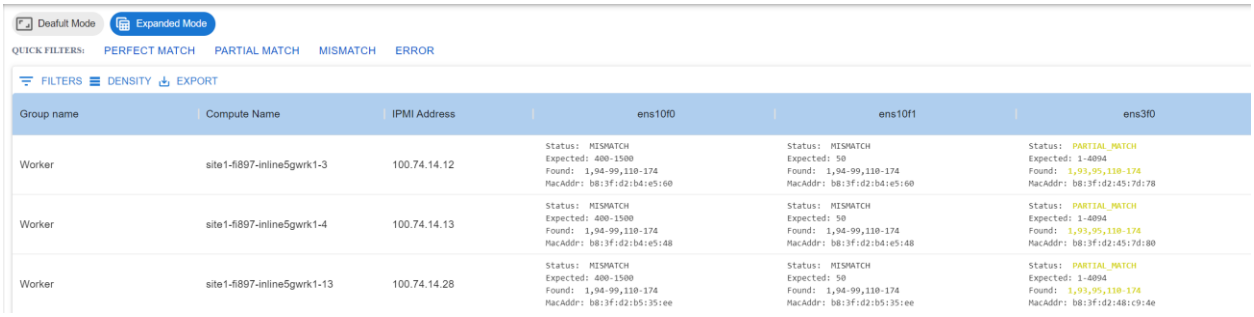
Interface Statistics

Expected Vlan Range	Match Vlan Range	Result ↑
1-200	1,94-99,110-174	PARTIAL MATCH

Also for each interface we can have more details, etc mac address

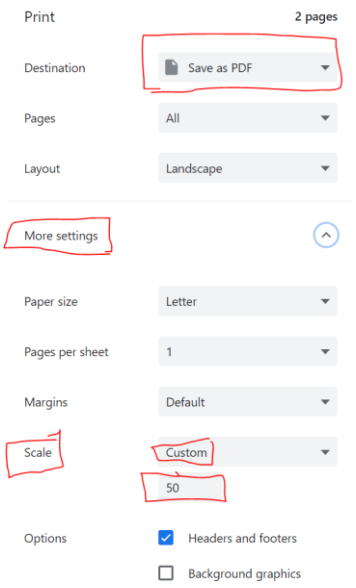

```
{
  "expectedVlanRange" : "1-200"
  "finalVlanMatch" : "PARTIAL_MATCH"
  "fw_version" : "5.7-1.0.2"
  "mac_address" : "b8:3f:d2:b5:1a:dc"
  "vlanMatch" : {...}
  "vlanRange" : "1, 94-99, 110-174"
}
```

We also have expanded mode, to show all the data on the table



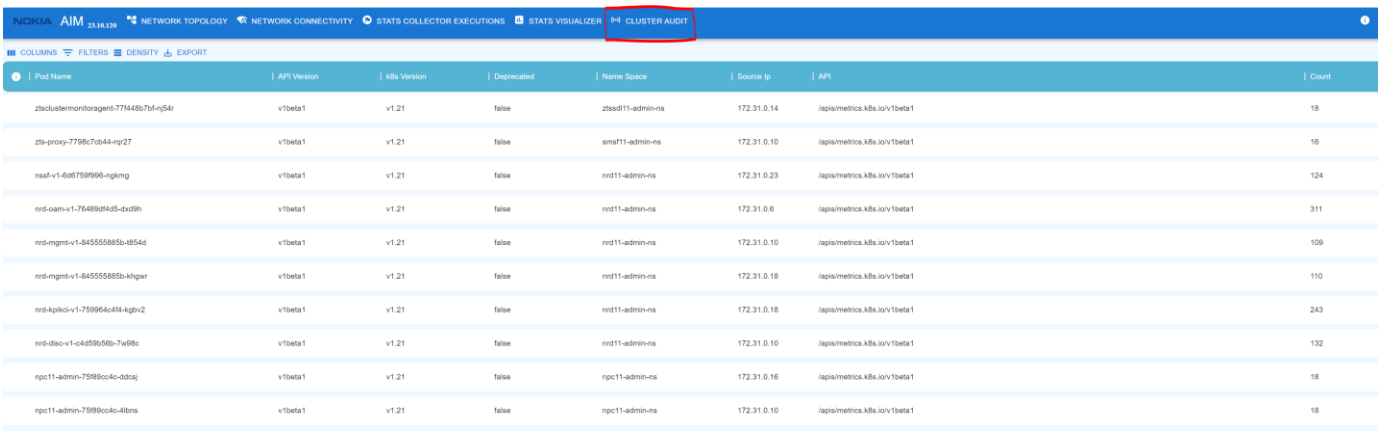
You can also export the report by using the export button:

You can export to pdf and change the scale of the table, so you get all the table in the pdf



5. Cluster Audit

This feature is for NCS



Pod Name	API Version	k8s Version	Deprecated	Name Space	Source Ip	API	Count
ztsclustermonitoragent-77f448b7bf-nj54r	v1beta1	v1.21	false	ztssdl11-admin-ns	172.31.0.14	/apis/metrics.k8s.io/v1beta1	18
zts-proxy-7798c7cb44-rqr27	v1beta1	v1.21	false	smsf11-admin-ns	172.31.0.10	/apis/metrics.k8s.io/v1beta1	16
nssf-v1-6d675ff996-ngkmg	v1beta1	v1.21	false	nrd11-admin-ns	172.31.0.23	/apis/metrics.k8s.io/v1beta1	124
nrd-oam-v1-76489df4d5-dxd9h	v1beta1	v1.21	false	nrd11-admin-ns	172.31.0.6	/apis/metrics.k8s.io/v1beta1	311
nrd-mgmt-v1-845555885b-t854d	v1beta1	v1.21	false	nrd11-admin-ns	172.31.0.10	/apis/metrics.k8s.io/v1beta1	109
nrd-mgmt-v1-845555885b-khgwr	v1beta1	v1.21	false	nrd11-admin-ns	172.31.0.18	/apis/metrics.k8s.io/v1beta1	110
nrd-kpikci-v1-759964c4f4-kgbv2	v1beta1	v1.21	false	nrd11-admin-ns	172.31.0.18	/apis/metrics.k8s.io/v1beta1	243
nrd-disc-v1-c4d59b56b-7w98c	v1beta1	v1.21	false	nrd11-admin-ns	172.31.0.10	/apis/metrics.k8s.io/v1beta1	132
npc11-admin-75f89cc4c-ddcsj	v1beta1	v1.21	false	npc11-admin-ns	172.31.0.16	/apis/metrics.k8s.io/v1beta1	18
npc11-admin-75f89cc4c-4lbns	v1beta1	v1.21	false	npc11-admin-ns	172.31.0.10	/apis/metrics.k8s.io/v1beta1	18

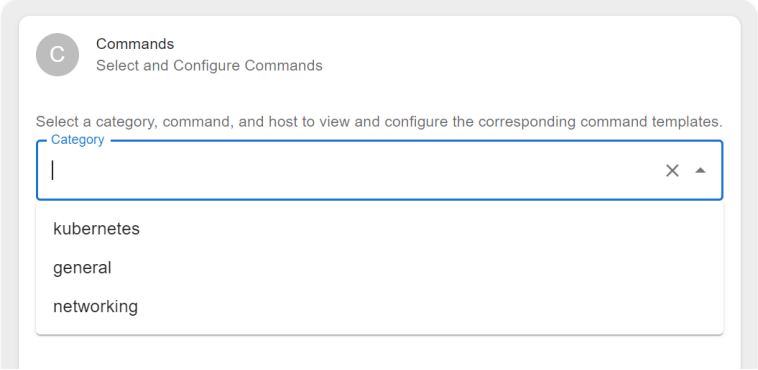
- The API audit logs the utilization of APIs across various applications and pods, providing a comprehensive overview of potentially problematic API usage. This includes highlighting APIs that are deprecated and slated for removal in upcoming releases, as well as those utilizing the alpha and beta versions of Kubernetes v1. By doing so, users are proactively notified and cautioned, empowering them to initiate necessary modifications in forthcoming versions to ensure seamless functionality and compliance
- It shows the number of how much a problematic API has been used by an application
- If the API is deprecated and what release of k8s it will be removed
- The type of the api, is it alpha,beta, …
- We recommend all apps who use these apis to use the updated version, since these apis will be removed in the future

6. Command Ease

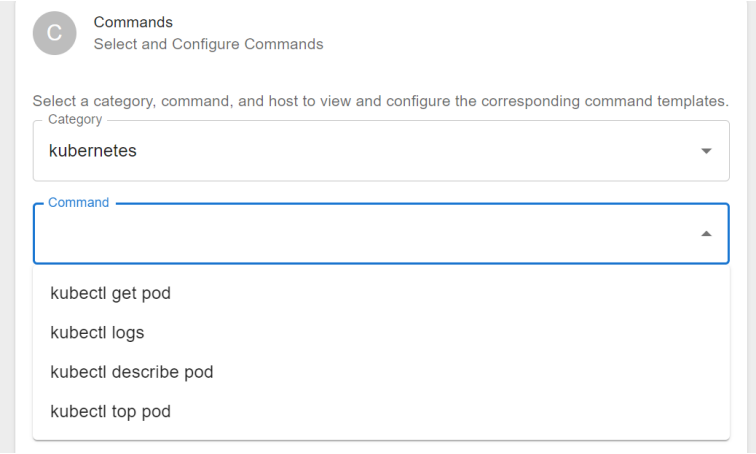
This feature is for NCS

Command Ease is a feature that gives the user the ability to running commands in an easy and convenient way which dosn't require SSH access to the cluster, the allowed commands are 'READ ONLY' commands.

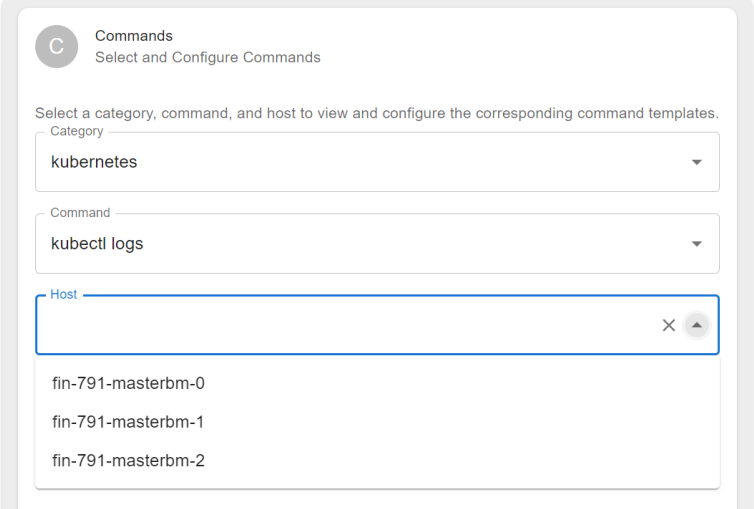
First you choose the command category:



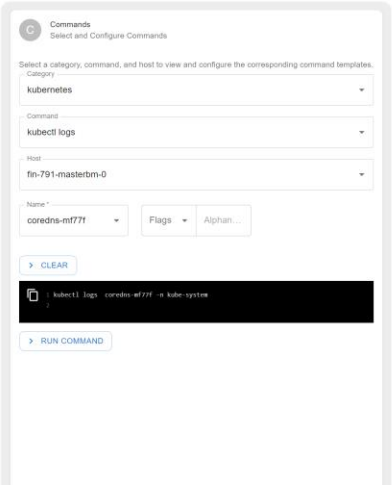
Then you choose the available command:

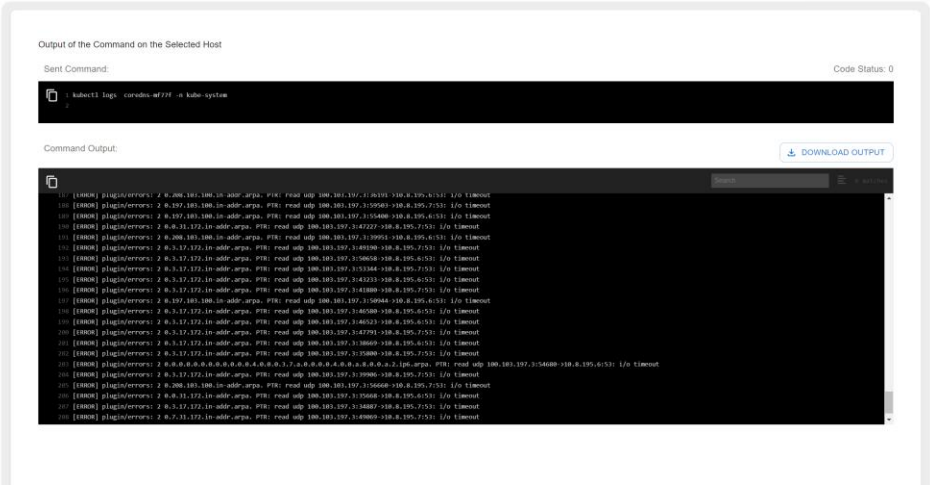


Then on which Node you want to run the command:



Then you choose the parameters for the command, and you get your output:





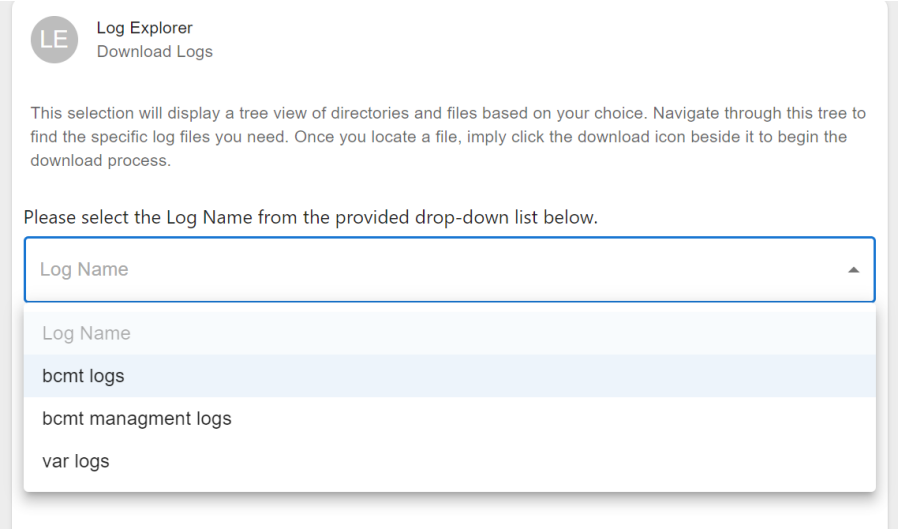
7. Log Explorer

This feature is for NCS

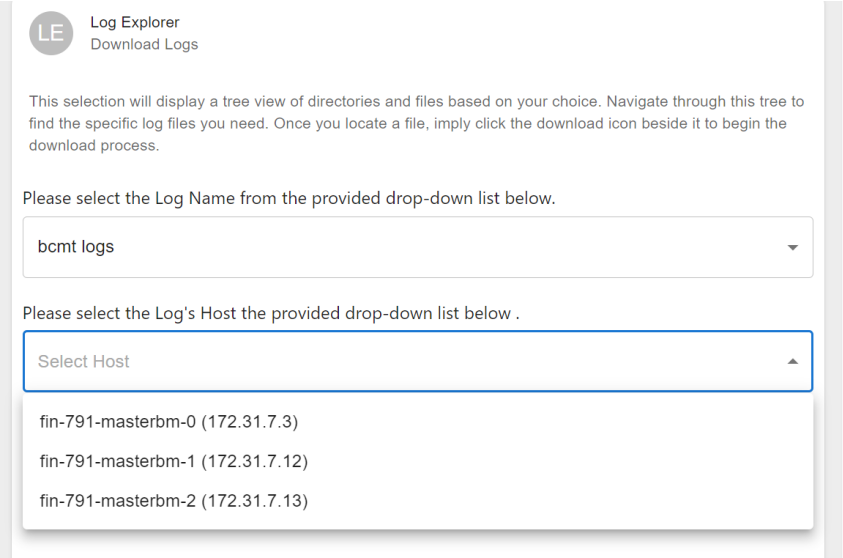
This Feature gives a tree view of directories and files based on your choice.

Navigate through this tree to find the specific log files you need. Once you locate a file, imply click the download icon beside it to begin the download process.

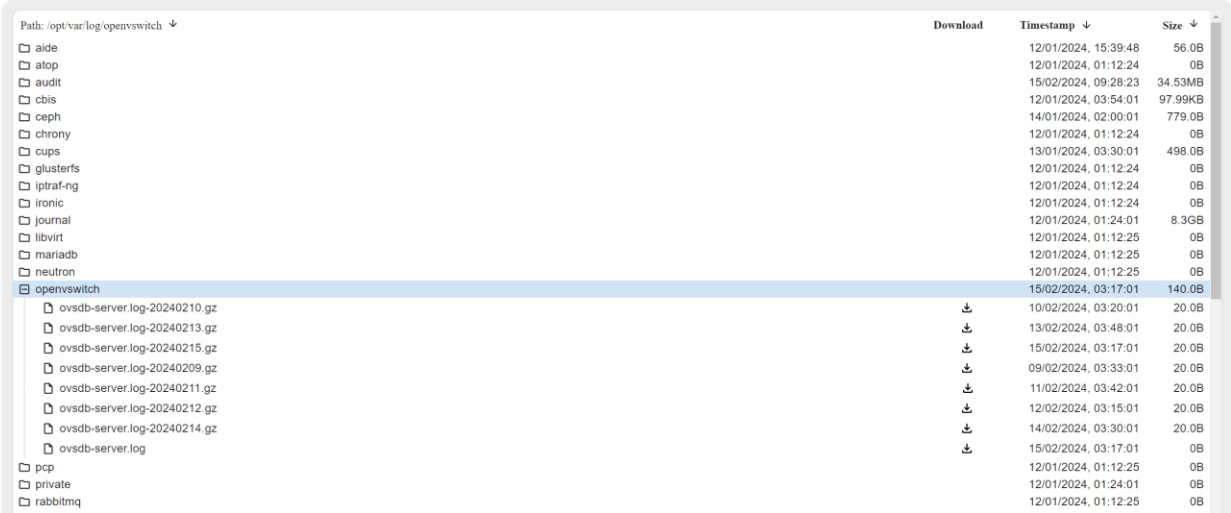
First you choose the kind of logs you need:



Then on which node you want:



Then you will get the directories tree:

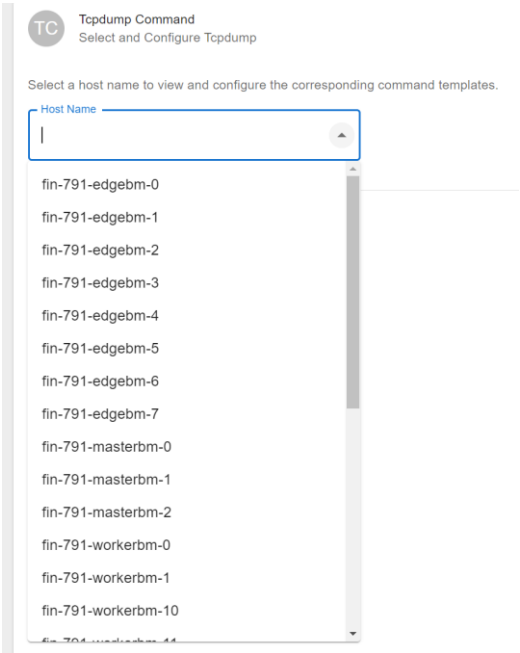


Path: /opt/var/log/openvswitch	Download	Timestamp	Size
aide		12/01/2024, 15:39:48	56.0B
atop		12/01/2024, 01:12:24	0B
audit		15/02/2024, 09:28:23	34.53MB
cbis		12/01/2024, 03:54:01	97.99KB
ceph		14/01/2024, 02:00:01	779.0B
chrony		12/01/2024, 01:12:24	0B
cups		13/01/2024, 03:30:01	498.0B
glusterfs		12/01/2024, 01:12:24	0B
iptraf-ng		12/01/2024, 01:12:24	0B
ironic		12/01/2024, 01:12:24	0B
journal		12/01/2024, 01:24:01	8.3GB
libvirt		12/01/2024, 01:12:25	0B
mariadb		12/01/2024, 01:12:25	0B
neutron		12/01/2024, 01:12:25	0B
openvswitch		15/02/2024, 03:17:01	140.0B
ovsdb-server.log-20240210.gz		10/02/2024, 03:20:01	20.0B
ovsdb-server.log-20240213.gz		13/02/2024, 03:48:01	20.0B
ovsdb-server.log-20240215.gz		15/02/2024, 03:17:01	20.0B
ovsdb-server.log-20240209.gz		09/02/2024, 03:33:01	20.0B
ovsdb-server.log-20240211.gz		11/02/2024, 03:42:01	20.0B
ovsdb-server.log-20240212.gz		12/02/2024, 03:15:01	20.0B
ovsdb-server.log-20240214.gz		14/02/2024, 03:30:01	20.0B
ovsdb-server.log		15/02/2024, 03:17:01	0B
pcp		12/01/2024, 01:12:25	0B
private		12/01/2024, 01:24:01	0B
rabbitmq		12/01/2024, 01:12:25	0B

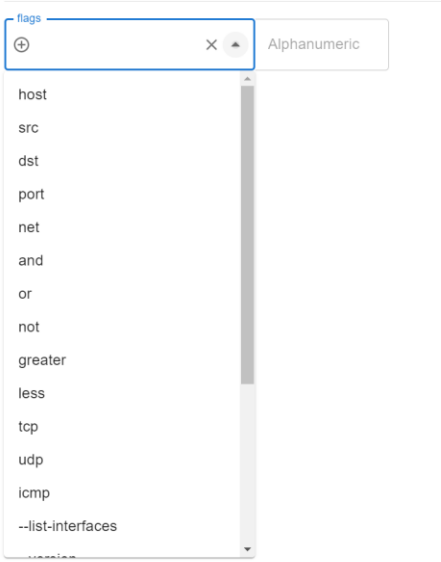
8. Tcpdump

Execute tcpdump commands via AIM on selected servers for packet capture, with options for time limit or file size restriction. Captured data streams directly to the browser, avoiding file storage on the cluster

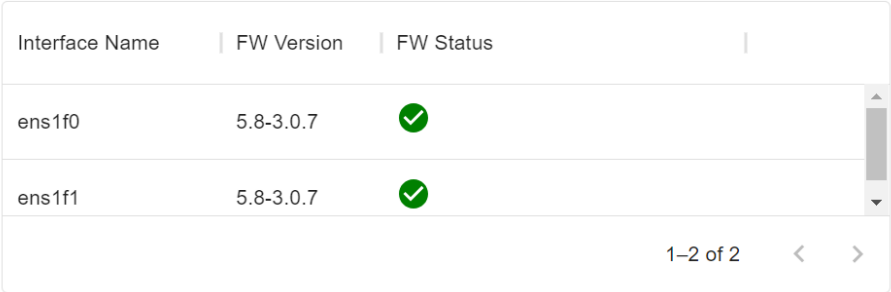
First you choose on what node you want to run tcpdump:



Then you start adding flags to the command:



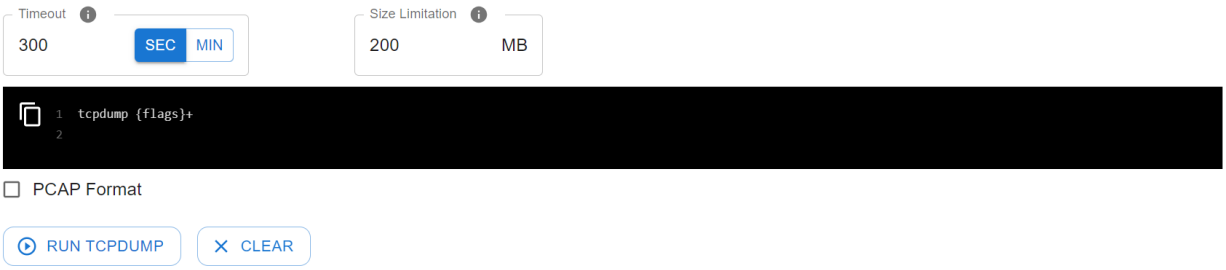
We will show you the current physical interfaces on the selected node you chose and its firmware version(some firmware version has bugs with tcpdump so we don't allow running it on these versions):



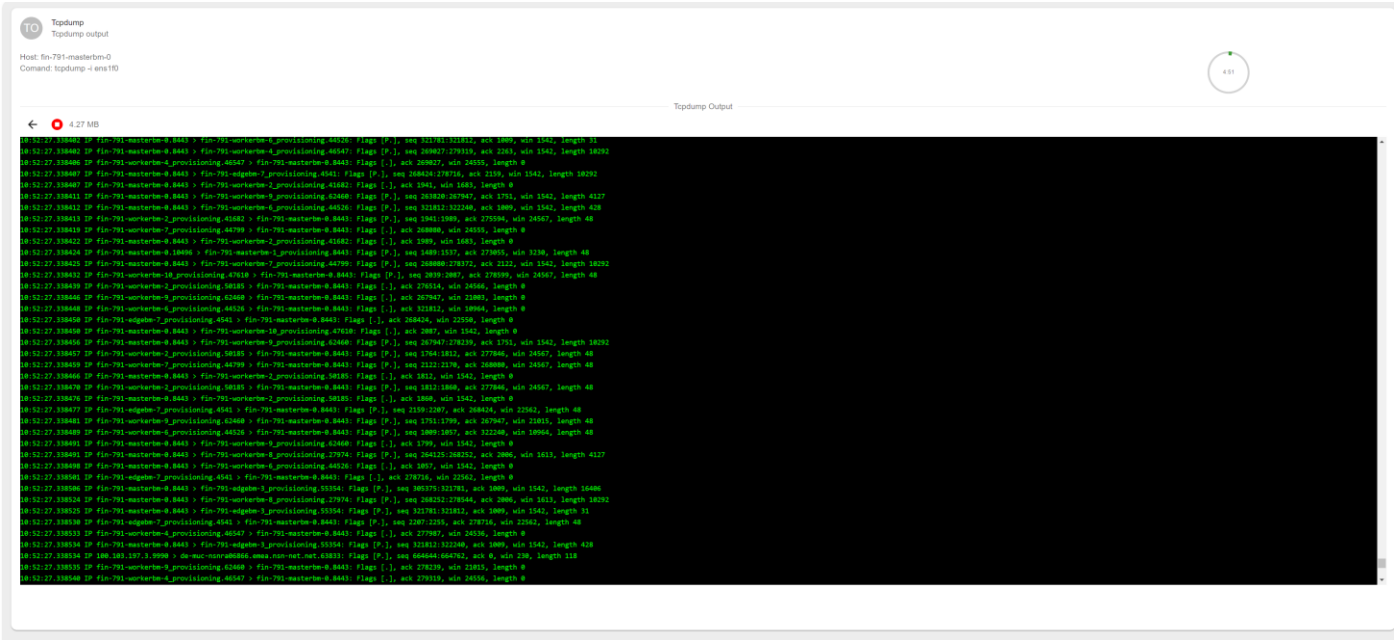
Interface Name	FW Version	FW Status
ens1f0	5.8-3.0.7	✓
ens1f1	5.8-3.0.7	✓

1–2 of 2

1- We do not allow to run the command indefinitely so you must choose a timeout.
2- Also the data will be transferred to the browser and not stay on the cluster, so we give a size limitation, this can be changed up to 500mb
3- You can choose PCAP file format, so you will get data in bytes and download it, then use wireshark or any packet analyzer tool to analyze your data



Here is an example of a run:

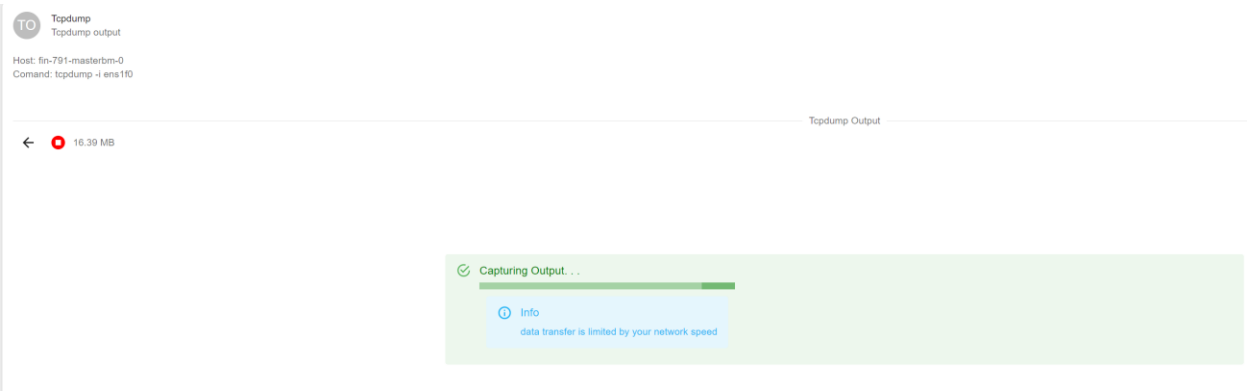


You can stop the command any time you like.

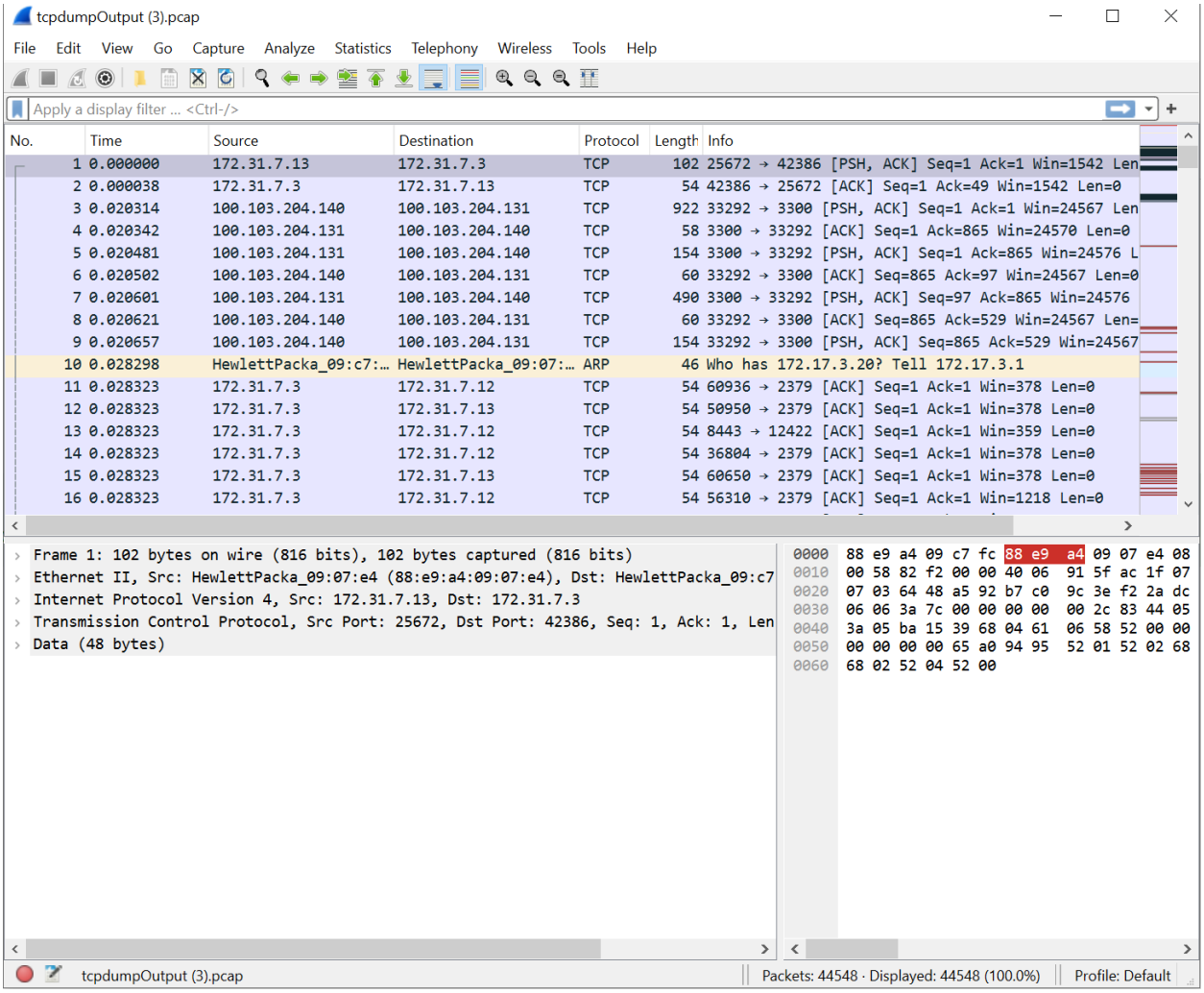
There is a timer when the command will finish(for the time out)

After stopping or after the timeout finish, you can download your data:

Here is an example of pcap file format:



After downloading it and uploading it to Wireshark:



9. Known issues and limitations

- Currently supporting CBIS platform (CBIS 19+).
- Currently supporting NCS platform (version 20+ and above).
- Currently only 1 sample can be active at any time.

10. Jira project

To open tickets about AIM, please use the following JIRA project:

- https://jiradc2.ext.net.nokia.com
    - Project: NFVI
    - Component: AIM

11. Service Policy and support

The AIM team can be contacted for support during the working hours and using the methods listed below:

Sundays – Thursdays, 08:00 – 18:00 (IST)

Omri.sabach@nokia.com

Nir.rashti@nokia.com

Odai.odeh@nokia.com